

Liouville signs at almost-prime pairs: a reduction to one inequality and two model sequences

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Abstract

Let λ be the Liouville function and $h \geq 2$ a fixed even integer. For fixed $c > 0$, put $\sigma = c/\log X$, $J_\sigma(n) = \prod_{p|n}(1 - p^{-\sigma})$ and $W_X(n) = J_\sigma(n)J_\sigma(n+h)$. We prove that $M_0 = \sum_{n \leq X} W_X(n) \asymp_{h,c} X/\log^2 X$, and that the mass with $\Omega(n) > K$ or $\Omega(n+h) > K$ is at most $C_{h,c}u^{-K}X/\log^2 X$, where $u = 2^{1/1000}$. For $0 < \varepsilon \leq 1/4$ and fixed $c \geq c_0(\varepsilon)$, the sums $\sum_{n \leq X} \lambda(n)W_X(n)$ and $\sum_{n \leq X} \lambda(n+h)W_X(n)$ have absolute values of total at most εM_0 for all large X . Consequently the inequality $|\sum_{n \leq X} \lambda(n)\lambda(n+h)W_X(n)| \leq (1-2\varepsilon)M_0$, for all large X , would give $\gg_{h,c,\varepsilon} X/\log^2 X$ occurrences of each sign pattern with $\Omega(n), \Omega(n+h) \leq K$ for some fixed K . We do not prove this inequality. We construct a sign sequence with square-root bounds, up to $\sqrt{\log X}$, on intervals, progressions and additive twists, on character twists of polynomial modulus and height, on fixed nonproportional positive-slope affine pairs and on reflected pairs. It has small root remainders at every fixed level $\exp((\log X)^\alpha)$, $0 < \alpha < 1$, but its root remainder for $(n, n+2)$ at X^θ is asymptotic to $(c_2 \log 2/32)\theta^2 X \log X$ for every fixed $0 < \theta < 1$. For every fixed nonzero h we also construct a sequence with the stated affine, character and Type I bounds whose matrices $(f(m\ell + h))$ have operator norm $\gg_h \sqrt{MN}/\log X$ on infinitely many balanced rectangles with $MN \asymp X$. One sequence gives these lower bounds for $h = 1, 2$ on alternating scales. These constructions show that the respective root-remainder and Type II targets require input beyond these listed numerical bounds. We classify exact separated-variable substitutions of polynomial height and bound their piece count. Finally, for a defined class of nonnegative divisor weights in the independent divisibility law, an individual-root positive second-moment normalisation forces $v \leq \exp(O_\delta(L))$ when the padding primes are at most e^L ; it therefore does not extend to $v = X^\theta$ with $X = \exp(L^{625})$.

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1 Introduction

OpenAI's proof of the two-point Chowla conjecture [20] (family 007 of the release, pinned to repository commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`) is the starting point of [12], whose proofs adapt its graph argument. For fixed even $h \geq 2$, [12, Theorem 1.1] gives, uniformly for $2 \leq z \leq \exp((\log X)^{1/3750})$, the count $\frac{1}{4}XV_h(z)(1+o(1))$ for each sign pattern of $(\lambda(n), \lambda(n+h))$ among the $n \leq X$ for which n and $n+h$ have no prime factor up to z ; here $V_h(z)$ is the sieve density of Section 2. By [12, Theorem 1.2], each pattern occurs for $\gg_h X/(\log X)^{1/1875}$ integers $n \leq X$ such that n and $n+h$ are squarefree, have no prime factor up to $\exp((\log X)^{1/3750})$, and satisfy $\Omega(n), \Omega(n+h) \leq (1 - 1/4000) \log \log X$. Here we ask what additional information would give a bounded number of prime factors. Chen's theorem gives infinitely many primes p with $p+2$

a prime or a product of at most two primes [3]. The parity problem concerns the extent to which divisibility information alone distinguishes integers with an even number of prime factors from those with an odd number; Selberg's work and Bombieri's asymptotic sieve are classical points of reference [2, 9, 22]. Goldston, Graham, Pintz and Yıldırım [10] obtain, in particular, infinitely many consecutive integers both having $\Omega = 5$. Matomäki, Radziwiłł and Tao [16] prove positive lower natural density for each of the eight consecutive three-term Liouville sign patterns. These statements concern different restrictions on prime factors and signs.

Sequences that satisfy sieve hypotheses and violate a conclusion are a classical tool in this subject. By Selberg's examples and Bombieri's work there are sequences satisfying Type I estimates at every level $\gamma < 1$ and vanishing on the primes; see [7, Section 1]. Ford and Maynard [7] give a general procedure for constructing non-negative sequences that satisfy prescribed Type I and Type II estimates, and show that a substantial Type II range is always necessary to guarantee a non-trivial lower bound for the sum over primes. Friedlander and Iwaniec [8] add an axiom to the classical formulation of the sieve which breaks the parity problem and detects primes in thin sequences. Tao [23] formulates a parity obstruction for properties defined by sign patterns of λ , presuming a Liouville pseudorandomness conjecture. Deligiannis [4] deduces a lower bound of the expected order for the number of Goldbach representations from two cancellation hypotheses for sifted two-point Liouville sums, and [5] analyses limitations of existing methods for such cancellation; Arneeth [1] formulates a sifted fixed-shift Liouville cancellation as the remaining analytic problem of a programme on the binary Goldbach problem. The sequences of Theorems 1.4 and 1.5 below are ± 1 -valued sequences for two-point sums at a fixed shift, with the lists of estimates stated there.

All logarithms are natural. We write $e(t) = \exp(2\pi it)$, $\lambda(n) = (-1)^{\Omega(n)}$, where Ω counts prime factors with multiplicity, and $\omega(n)$ for the number of distinct prime factors. The divisor function is τ , the Möbius function is μ , and $P^-(1) = \infty$. Cutoffs in sums mean integer floors, and only positive arguments of f or λ are included. Constants in O and $\ll$ are independent of the growing variable unless indicated. Define

$$\begin{aligned} \sigma &= c/\log X, & J_\sigma(n) &= \prod_{p|n} (1 - p^{-\sigma}), & W_X(n) &= J_\sigma(n)J_\sigma(n+h), \\ M_0 &= \sum_{n \leq X} W_X(n), & M_1 &= \sum_{n \leq X} \lambda(n)W_X(n), & M_2 &= \sum_{n \leq X} \lambda(n+h)W_X(n), \\ M_{12} &= \sum_{n \leq X} \lambda(n)\lambda(n+h)W_X(n). \end{aligned} \tag{1.1}$$

Theorem 1.1 (Jordan weights). *Fix even $h \geq 2$ and $c > 0$. There are positive $a_{h,c}, C_{h,c}, C'_{h,c}$ such that, for all sufficiently large X ,*

$$\begin{aligned} a_{h,c} \frac{X}{\log^2 X} &\leq M_0 \leq C_{h,c} \frac{X}{\log^2 X}, \\ \sum_{n \leq X} W_X(n) \{u^{\Omega(n)} + u^{\Omega(n+h)}\} &\leq C'_{h,c} \frac{X}{\log^2 X}, \quad u = 2^{1/1000}. \end{aligned} \tag{1.2}$$

For every $K \geq 0$, uniformly in X, K in this range,

$$\sum_{\substack{n \leq X \\ \Omega(n) > K \text{ or } \Omega(n+h) > K}} W_X(n) \leq C'_{h,c} u^{-K} \frac{X}{\log^2 X}. \tag{1.3}$$

For every $0 < \varepsilon \leq 1/4$ there is $c_0(\varepsilon) > 0$, independent of fixed even h , such that for fixed $c \geq c_0(\varepsilon)$ and all sufficiently large X ,

$$|M_1| + |M_2| \leq \varepsilon M_0. \tag{1.4}$$

The threshold may depend on h, c, ε .

Theorem 1.2 (Reduction to one inequality). *Fix $0 < \varepsilon \leq 1/4$, $c \geq c_0(\varepsilon)$ and even $h \geq 2$. Suppose that for every sufficiently large X ,*

$$|M_{12}| \leq (1 - 2\varepsilon)M_0. \quad (1.5)$$

Each $(s, t) \in \{\pm 1\}^2$ then has weighted mass at least $\varepsilon M_0/4$. Choose a fixed integer K with $C'_{h,c} u^{-K} \leq \varepsilon a_{h,c}/8$. For all sufficiently large X , the count of this pattern with $\Omega(n), \Omega(n+h) \leq K$ is at least $\varepsilon a_{h,c} X / (8 \log^2 X)$. In particular K is independent of X .

For any sign sequence and fixed positive shift h put

$$\begin{aligned} S_f^{(h)}(X; d, r) &= \sum_{\substack{n \leq X \\ n \equiv r \pmod{d}}} f(n) f(n+h), \quad \mathcal{R}_h(d) = \{r \pmod{d} : r(r+h) \equiv 0 \pmod{d}\}, \\ R_f^{(h)}(X, D) &= \sum_{d \leq D} \mu^2(d) \sum_{r \in \mathcal{R}_h(d)} |S_f^{(h)}(X; d, r)|. \end{aligned} \quad (1.6)$$

For $h = 2$ omit the superscript.

Proposition 1.3 (Consumers of root remainder bounds). *Fix even $h \geq 2$. Suppose fixed $\theta > 0$, $C_{\text{rem}} < \infty$ and X_0 satisfy*

$$R_\lambda^{(h)}(X, X^\theta) \leq C_{\text{rem}} X / \log^2 X \quad (X \geq X_0). \quad (1.7)$$

The sum includes every root in the support of the sieve weights. Then fixed $K, c_1 > 0$ and X_1 exist, depending on $h, \theta, C_{\text{rem}}$ and the classical input constants, such that each ordered sign pattern occurs at least $c_1 X / \log^2 X$ times with both Ω values at most K , for $X \geq X_1$. Equivalently, the fixed- K count is

$$\#\{1 \leq n \leq X : \Omega(n), \Omega(n+h) \leq K, \lambda(n) = s, \lambda(n+h) = t\} \geq c_1 X / \log^2 X. \quad (1.8)$$

Only finiteness of C_{rem} is required. The reflected conclusion for every large even N requires the separate hypothesis

$$\sum_{d \leq N^\theta} \mu^2(d) \sum_{\substack{r(N-r) \equiv 0 \pmod{d}}} \left| \sum_{\substack{1 \leq n < N \\ n \equiv r \pmod{d}}} \lambda(n) \lambda(N-n) \right| \leq C_{\text{rem}} N / \log^2 N, \quad (1.9)$$

with fixed constants uniform in N and all these roots; it gives fixed K, c_1, N_1 and the lower bound $c_1 N / \log^2 N$ for every ordered pattern and every even $N \geq N_1$. For the smaller level, fix $0 < \alpha < 1$ and put $D_\alpha(X) = \exp((\log X)^\alpha)$. Suppose a fixed $C < \infty$ satisfies, for every sufficiently large X ,

$$\sum_{d \leq D_\alpha(X)} \mu^2(d) \sum_{\substack{r \pmod{d} \\ r(r+h) \equiv 0 \pmod{d}}} \left| \sum_{\substack{n \leq X \\ n \equiv r \pmod{d}}} \lambda(n) \lambda(n+h) \right| \leq C X / \log^2 X. \quad (1.10)$$

Thus every squarefree modulus in this range and every root is included. Then, for each sufficiently large fixed s_0 , every pattern occurs at least $c_{h,s_0} X / (\log X)^{2\alpha}$ times on the support $P^-(n), P^-(n+h) > \exp((\log X)^\alpha / s_0)$, with

$$\Omega(n), \Omega(n+h) \leq s_0 (\log X)^{1-\alpha} + O_h(1). \quad (1.11)$$

Constants may also depend on the fixed α . The reflected version requires its own bound with squarefree $d \leq \exp((\log N)^\alpha)$ and every root $r(N-r) \equiv 0 \pmod{d}$, with a fixed constant uniform for all sufficiently large even N .

Theorem 1.4 (A sequence for the sieve remainder). *There are an absolute constant C and one $f : \mathbb{N} \rightarrow \{\pm 1\}$ with the following properties. For all sufficiently large integer M , all integer intervals $I \subset [1, M]$, $q \leq M$, $r \bmod q$ and real β ,*

$$\left| \sum_{\substack{n \in I \\ n \equiv r \pmod{q}}} f(n) e(\beta n) \right| \leq C \sqrt{t \log(2M)}, \quad t = |I \cap (r + q\mathbb{Z})|. \quad (1.12)$$

For each fixed $B > 0$, eventually (with a threshold allowed to depend on B), uniformly over $q \leq M^B$, all $\chi \bmod q$, $|v| \leq M^B$ and these intervals,

$$\left| \sum_{n \in I} f(n) \chi(n) n^{iv} \right| \leq C_B \sqrt{|I| \log(2M)}. \quad (1.13)$$

Every fixed nonproportional positive-slope integer affine pair, with fixed progression restrictions allowed, has correlation $O_{a_1, a_2, b_1, b_2, f}(\sqrt{X \log(2X)})$. The same bound holds uniformly for nonzero determinant and $1 \leq a_i \leq (\log X)^B$, $0 \leq b_i \leq (\log X)^B$, with constant depending only on B, f . Also

$$\begin{aligned} \sup_{1 \leq h \leq X} \left| \sum_{n \leq X} f(n) f(n+h) \right| &\ll_f \sqrt{X \log(2X)}, \\ \left| \sum_{1 \leq n < N} f(n) f(N-n) \right| &\ll_f \sqrt{N \log(2N)} \quad (N \geq 2). \end{aligned} \quad (1.14)$$

Define $A_f(X, D) = \sum_{d \leq D} d^{-1} \sum_{r \bmod d} |S_f(X; d, r)|$ over all moduli. For every fixed $0 < \theta < 1$, eventually

$$A_f(X, X^\theta) \ll_\theta X \exp\left(-\frac{\log X}{4 \log \log X}\right), \quad (1.15)$$

whereas

$$R_f(X, X^\theta) \sim \frac{c_2 \log 2}{32} \theta^2 X \log X, \quad c_2 = \frac{3}{8} \prod_{p > 2} (1 + 2/p)(1 - 1/p)^2 > 0. \quad (1.16)$$

For every fixed $0 < \alpha < 1$, uniformly for $2 \leq D \leq \exp((\log X)^\alpha)$,

$$R_f(X, D) \leq X^{1/2 + o_\alpha(1)}. \quad (1.17)$$

The sequence may be chosen not completely multiplicative. All fixed real parameters here share one sequence.

Theorem 1.5 (A sequence for Type II sums). *For every fixed integer $h \neq 0$ there is one $f : \mathbb{N} \rightarrow \{\pm 1\}$ and constants $C_h, Y_h > 0$ such that, for integer $Y \geq Y_h$, all nonzero integer a, c , all integers b, d with $|a|, |b|, |c|, |d| \leq Y$, and integer intervals $I \subset [1, Y]$ on which the arguments lie in $[1, Y]$,*

$$\begin{aligned} \left| \sum_{n \in I} f(an + b) \right| &\leq C_h Y^{3/4} \sqrt{\log(2Y)}, \\ \left| \sum_{n \in I} f(an + b) f(cn + d) \right| &\leq 1 + C_h Y^{3/4} \sqrt{\log(2Y)} \quad ((a, b) \neq (c, d)). \end{aligned} \quad (1.18)$$

For each fixed $A > 0$, all $q \leq x^A$ and $\chi \bmod q$ satisfy $\sum_{n \leq x} f(n)\chi(n) \ll_{h,A} x^{3/4} \sqrt{\log(2x)}$. For fixed $0 \leq \tau < 1/7$,

$$\sum_{n \leq L} f(qn + r) \ll_{h,\tau} L^{1-(1-7\tau)/16}, \quad 1 \leq q \leq L^\tau, \quad 1 \leq r \leq q. \quad (1.19)$$

There are $c_h > 0$ and $T_j \rightarrow \infty$ such that, on $A_j = (T_j, 2T_j] \cap \mathbb{Z}$ and $B_j = (3T_j, 6T_j] \cap \mathbb{Z}$,

$$\|(f(m\ell + h))_{A_j \times B_j}\|_{\text{op}} \geq c_h T_j / \log T_j \asymp_h \sqrt{|A_j||B_j|/\log(T_j^2)}. \quad (1.20)$$

The comparison on the right concerns the lower-bound scale. A single sequence can have these properties and infinitely many such rectangles for $h = 1$ on even scales and $h = 2$ on odd scales.

An exact separated-variable substitution on $A \times B$ means maps into positive integers and nonzero rational row and column weights satisfying

$$U(x)V(y) + h = R(x)S(y)(xy + h). \quad (1.21)$$

Such a substitution on a product set $A \times B$ is called a piece; it preserves products if $U(x)V(y) = xy$ on $A \times B$.

Theorem 1.6 (Classification and piece count). *For finite $A, B \subset \mathbb{Q}$, each with at least two elements, and nonzero rational h , every rational-valued substitution (1.21) has U, V injective, and there are $a, b, c, d \in \mathbb{Q}$ with $\Delta = ad - bc \neq 0$ such that*

$$U(x) = \frac{ax + b}{cx + d}, \quad V(y) = \frac{h(dy - hc)}{ha - by}, \quad R(x)S(y) = \frac{h\Delta}{(cx + d)(ha - by)}, \quad (1.22)$$

with nonzero denominators. Conversely these formulas give the identity. Fix integer $h \neq 0$, $C \geq 1$ and $\eta > 0$. There is $K_{h,C,\eta} > 0$ such that for all $X \geq 2$, $A, B \subset [1, X] \cap \mathbb{Z}$ and every piece on $A \times B$ with U, V valued in $[1, X^C] \cap \mathbb{Z}$, either the piece preserves products or $\min(|A|, |B|) \leq K_{h,C,\eta} X^\eta$. If Q such pieces that do not preserve products cover at least ρ prs points of $A_0 \times B_0$, where $A_0, B_0 \subset [1, X] \cap \mathbb{Z}$, $r = |A_0| \geq 1$, $s = |B_0| \geq 1$ and $0 < \rho \leq 1$, then

$$Q \geq \frac{\rho \min(r, s)}{K_{h,C,\eta} X^\eta}. \quad (1.23)$$

For fixed h and large X , if $xy + h > 0$ throughout $A_0 \times B_0$, a covering by $\min(r, s)$ such pieces exists with outputs at most X^3 . When $r, s \asymp X^\delta$ with fixed $\delta > 0$ and $xy + h > 0$ on $A_0 \times B_0$, the minimum number of pieces that do not preserve products, have outputs at most X^3 and cover $A_0 \times B_0$ lies between $X^{\delta-o(1)}$ and $O(X^\delta)$.

The class $\mathcal{C}_+$. Let $\mathcal{P}$ be a finite set of primes at most $Y = e^L \geq 2$, and let $a(u) \geq 0$ be a finite-support weight on $\mathcal{P}$ -smooth integers, depending only on u , with $a(1) > 0$; a need not be multiplicative. In the independent Haar p -adic law put $w(n) = \sum_{u|n} a(u)$ and $S = \sum_u a(u)/u$. Tilt the law by w/S , and choose U with the original conditional probabilities $q_n(u) = a(u)\mathbf{1}_{u|n}/w(n)$. Add an independent uniform coordinate modulo a prime $v \notin \mathcal{P}$. For fixed integers h, d impose $v \mid n + hdu$. A measurable retention indicator $B(n, u) \in \{0, 1\}$ may read the vertex, this extra residue coordinate and additional independent coordinates, including independent random coins, but must retain mass

$$\mathbb{E} \left[w \sum_u q_n(u) \mathbf{1}_{B(n,u)} \mathbf{1}_{v|n+hdu} \right] \geq \delta S/v, \quad 0 < \delta \leq 1 \quad \text{fixed}. \quad (1.24)$$

Assign every retained (n, u) to exactly one bin, and define $\rho_{j,v}$ by summing the original $q_n(u)$ with the retention and target indicators in that bin. The class consists of schemes requiring

$$\mathbb{E} \left[w \sum_j \rho_{j,v}^2 \right] \leq \frac{CS}{v^2 L} (\log L)^\beta \quad (1.25)$$

for fixed C, β , as $L \rightarrow \infty$. Retention is an indicator on this enlarged probability space. Replacing it inside the row sum by a coefficient $b(n, u)$ taking values strictly between zero and one is outside $\mathcal{C}_+$.

Theorem 1.7 (Nonnegative divisor weights). *Every scheme in $\mathcal{C}_+$ necessarily satisfies*

$$\begin{aligned} v &\leq \frac{C}{\delta L} (\log L)^\beta \exp(H_{\mathcal{P}}/\delta) \leq \exp(O_\delta(L)), \\ H_{\mathcal{P}} &= \sum_{p \in \mathcal{P}} \left[-\log(1 - 1/p) + \frac{\log p}{p-1} \right] \ll \log(2Y). \end{aligned} \quad (1.26)$$

Thus the required normalisation does not hold at $v = X^\theta$, $X = \exp(L^{625})$, for fixed $\theta > 0$. The implicit constant may depend on fixed C, β as well.

What the model sequences show and what they do not

Theorem 1.4 supplies precisely the interval, progression, additive-twist and polynomial-modulus and polynomial-height character bounds, the fixed affine bounds and their stated polylogarithmic extension, the all-shift and reflected bounds, the displayed normalised residue saving, and all fixed stretched-exponential root bounds. It also supplies the short-progression variance bound proved in Section 3. The constants in the eventual concentration bounds are absolute, or depend only on the displayed fixed parameters. The finite thresholds may depend on the realisation f and are not specified numerically; a hypothesis with a prescribed numerical starting point is outside this statement. The restriction to nonproportional affine pairs in the numerical input list is deliberate. For example,

$$\sum_{n \leq X} f(n)f(2n) = O(\sqrt{X \log(2X)}), \quad \lambda(2n) = -\lambda(n), \quad \sum_{n \leq X} \lambda(n)\lambda(2n) = -\lfloor X \rfloor. \quad (1.27)$$

For the model bound use (3.17) with ambient length $2X$ and (3.15): the distinct slopes leave at most three parameters with possibly nonzero mean. The Liouville identity uses complete multiplicativity and $\lambda(2) = -1$. An argument using only the listed numerical estimates, together with CRT, divisor-counting, sieve-weight identities and inequalities valid for arbitrary bounded sign sequences, applies equally to this f , whose positive-power root remainder exceeds $X/\log^2 X$ by an unbounded factor. A proof of $R_\lambda(X, X^\theta) \leq C_{\text{rem}} X/\log^2 X$ for λ , with a fixed $\theta > 0$ and a finite C_{rem} , must therefore use further information or stronger estimates than these specified strengths. Complete multiplicativity, the values at every prime, $\lambda = (-1)^\Omega$, Euler products, zero-free regions with uniform analytic control, and higher-order correlations are outside this list. The construction does not estimate the actual Liouville root remainder and does not address a fixed power saving for A_λ , parameters tending to the endpoints with X , or a computable generation algorithm for f . It can be chosen to fail complete multiplicativity explicitly.

Theorem 1.5 supplies the affine and reflected bounds at their displayed strengths, the Type I bound, the unweighted maximal residue and prefix bounds of Section 4, and the polynomial-modulus one-factor character means; one may also include the displayed growing-coefficient bound

there. A universal argument using only these data and finite algebra, norm inequalities, prime counts and sieve operations on the indices applies to its f . Its operator lower bound exceeds every fixed target $\sqrt{MN}(\log X)^{-3/2-\kappa}$, $\kappa > 0$, on infinitely many of the specified rectangles. The sequence fails the multiplicative law at every fixed prime. The construction does not supply actual Vaughan weights, all prime values, Euler identities, zero-free analytic inputs or higher Liouville correlations; it does not determine a global optimal operator upper bound or force failure on every large rectangle. Theorem 1.6 concerns Cartesian pieces, separate integer maps, exact identities, nonzero rational row and column weights and polynomial output height. Joint-variable maps, non-Cartesian masks, superpolynomial outputs and approximate identities are outside that statement.

Theorem 1.7 concerns precisely $\mathcal{C}_+$: nonnegative coefficients depending only on the divisor, the independent divisibility law, its original selection probabilities, fixed retained root mass, one bin per retained edge and the stated positive individual-root second moment. It allows nonmultiplicative coefficients and vertex-dependent retention. Signed padding, cancellation across roots or moduli, freely vertex-dependent coefficients $a(n, u)$, splitting or replicating one edge between bins, fractional row coefficients $b(n, u)$ taking values strictly between zero and one, and a mechanism not requiring (1.25) are outside the class. The retention indicator may depend on additional independent coordinates, but it still takes only the values zero and one; the nonnegative divisor coefficients $a(u)$ themselves may be nonmultiplicative. A direct Liouville pairing estimate is also outside it. No transfer of its independent-law lower bound to a finite interval of actual integers is asserted. The local centring and short-block statements below have their stated scopes and do not restrict every possible centring or long-block method.

The same sieve-model realisation has the rough-pair asymptotics at small levels described in Section 3, so its large-level root behaviour is compatible with [12]. The Type II model can also have both signs infinitely often at the shifted-prime positions $p + h$, in agreement with [13]. These are numerical compatibilities: the model signs are not $(-1)^\Omega$, and the multiplicative identities used for Liouville in [11–13] are additional inputs.

Section 2 proves the reduction and the two sieve consumers using classical inputs. Sections 3 and 4 construct the two sequences, and Section 5 proves the separated-variable classification and covering bound. Section 6 proves the positive-weight bounds and the related local and short-block statements. Remark 2.5 and Proposition 6.10 are relative to the zero-free half-plane $\Re s > 7/8$ of [19]; no other statement of this introduction uses it.

2 Jordan weights and the reduction

2.1 Classical sieve inputs

Put $P(z) = \prod_{p \leq z} p$, $\rho_h(d) = |\mathcal{R}_h(d)|$ on squarefree integers, $g_h(d) = \rho_h(d)/d$, and $V_h(z) = \prod_{p \leq z} (1 - g_h(p))$. For even h , $\rho_h(2) = 1$ and $\rho_h(p) \leq 2$ for odd primes; the factors are positive. The dimension-two product condition is uniform in h , since $\rho_h(p) \leq \rho_2(p)$.

Input 2.1 (Fundamental lemma). *For these densities, and for subsets of the forbidden roots, there are squarefree divisor weights $\xi_d^\pm$, supported on $d \mid P(z)$, $d \leq D$, with $|\xi_d^\pm| \leq 1$, such that*

$$w^-(n) \leq \mathbf{1}_{(n(n+h), P(z))=1} \leq w^+(n), \quad w^\pm(n) = \sum_{d \mid n(n+h)} \xi_d^\pm. \quad (2.1)$$

For sufficiently large $s = \log D / \log z$, with $2 \leq z \leq D^{1/2}$,

$$\left| \sum_d \xi_d^\pm g_h(d) - V_h(z) \right| \leq E(s) V_h(z), \quad E(s) \rightarrow 0. \quad (2.2)$$

We use the bounded-weight formulation of [6, Theorem 3.6(a), p. 38]; see also the standard fundamental-lemma treatment in [9]. Its error tends to zero, in fact exponentially, at fixed sieve dimension. The probability-space formulation allows arbitrary subsets of the two local roots. The constants are absolute for even h , including $h \leq 2X$, by the product condition just checked.

Lemma 2.2 (Maximal all-residue bound). *For fixed $0 < \eta < 1/2$ and every $A > 0$,*

$$\sum_{d \leq T^\eta} \max_{r \bmod d} \max_{y \leq T} \left| \sum_{\substack{n \leq y \\ n \equiv r \pmod{d}}} \lambda(n) \right| \ll_{A,\eta} T \log^{-A} T. \quad (2.3)$$

For $D \leq X^{1/4}$ and E_d the inner maximum at $T = 2X$,

$$\sum_{d \leq D} \mu^2(d) \rho_h(d) E_d \leq \left(\sum_{d \leq D} E_d \right)^{1/2} \left(\sum_{d \leq D} \mu^2(d) 4^{\omega(d)} E_d \right)^{1/2} \ll_B X \log^{-B} X \quad (2.4)$$

for every fixed $B > 0$. Nonunit classes are included.

Proof. The nonmaximal bound is [21, (1.5), p. 2], whose maximum is over all residue classes. For completeness, its nonunit range also follows from its reduced-class case $(a, k) = 1$. Write $E_{\text{red}}(Z; k) = \max_{(a,k)=1} |\sum_{m \leq Z, m \equiv a \pmod{k}} \lambda(m)|$. For a residue $r \pmod{d}$, let $g = (r, d)$ and put $n = gm$. Then $m \equiv r/g \pmod{d/g}$ is a reduced class, and complete multiplicativity gives $\lambda(n) = \lambda(g)\lambda(m)$; coprimality of g, m is not required. Choose $\eta < \eta' < 1/2$ and put $Q = T^\eta$. The maximum over residues is bounded by summing over the possible g , so

$$\begin{aligned} \sum_{d \leq Q} \max_{r \bmod d} \left| \sum_{\substack{n \leq T \\ n \equiv r \pmod{d}}} \lambda(n) \right| &\leq \sum_{g \leq Q} \sum_{k \leq Q/g} E_{\text{red}}(T/g; k) \\ &\ll_{H,\eta,\eta'} \sum_{g \leq Q} \frac{T/g}{\log^H(T/g)} \ll_{H,\eta} T \log^{1-H} T. \end{aligned} \quad (2.5)$$

Indeed $Q/g \leq (T/g)^{\eta'}$, since their ratio is $T^{\eta-\eta'} g^{\eta'-1} \leq 1$, and $T/g \geq T^{1-\eta}$, so the reduced-class theorem applies uniformly for every $g \leq Q$. Taking H sufficiently large gives any fixed logarithmic saving. This reduction specifically uses complete multiplicativity of λ . Fix $\eta < \eta' < 1/2$ and an integer $\ell > A + 3$. Choose $O(\log^\ell T)$ grid cutoffs spaced by at most $T/\log^\ell T$. At the nonzero cutoffs $y \geq T/(2\log^\ell T)$ we have $T^\eta \leq y^{\eta'}$ and $\log y \asymp \log T$, so (2.5) applies at the endpoint y to all moduli $d \leq T^\eta$ (with y in place of T and $Q = T^\eta$). Use saving $A + \ell + 3$ and sum over the grid; the cost is $O(T \log^{-A-2} T)$. Interpolation changes each progression by at most $T/(d \log^\ell T) + 1$. Its sum, including the small initial interval, is $O(T \log^{1-\ell} T + T^\eta)$, proving (2.3) with the maxima in the stated order. Since $\rho_h(d) \leq 2^{\omega(d)}$, Cauchy–Schwarz gives (2.4). The elementary bounds

$$\sum_{d \leq D} \frac{4^{\omega(d)}}{d} \leq (1 + \log D)^4, \quad \sum_{d \leq D} 4^{\omega(d)} \ll D(1 + \log D)^3 \quad (2.6)$$

follow from $4^{\omega(d)} \leq \tau_4(d)$ and four divisor variables. Thus $E_d \leq 2X/d + 1$ makes the second square $O(X \log^4(2X))$; arbitrary saving in (2.3) proves the assertion. For $\lambda(n + h)$ substitute $m = n + h$ and subtract two prefixes, once $X \geq h$. $\square$

CRT and $\rho_h(d) \leq \tau(d)$ give

$$\sum_{n \leq X} \mathbf{1}_{d|n(n+h)} = Xg_h(d) + O(\rho_h(d)), \quad \sum_{d \leq D} \mu^2(d)\rho_h(d) \ll D \log(2D). \quad (2.7)$$

Writing $W(n) = \mathbf{1}_{(n(n+h), P(z))=1}$, the fundamental lemma yields

$$\begin{aligned} \left| \sum_{n \leq X} w^\pm(n) - XV_h(z) \right| &\leq E(s)XV_h(z) + O(D \log(2D)), \\ \sum_{n \leq X} W(n) &= XV_h(z) + O(E(s)XV_h(z) + D \log(2D)), \\ \sum_{n \leq X} |w^+(n) - W(n)| &\leq 2E(s)XV_h(z) + O(D \log(2D)). \end{aligned} \quad (2.8)$$

The last bound uses $w^+ \geq W \geq w^-$ and is an L^1 comparison, valid after multiplication by any sign. Mertens' product theorem [17, Section 2.2] and a convergent positive product give an absolute $c_* > 0$ with

$$\begin{aligned} V_h(z) &\geq V_2(z) \geq c_*/\log^2(2z), \\ V_2(z) &= 2 \left\{ \prod_{p \leq z} (1 - 1/p) \right\}^2 \prod_{2 < p \leq z} \left(1 - \frac{1}{(p-1)^2} \right). \end{aligned} \quad (2.9)$$

The comparison is prime by prime, including 2.

Input 2.3 (Nair–Tenenbaum upper bound). *We use [14, Theorem 1] for the two linear polynomials $Q_1(t) = t$, $Q_2(t) = t + h$, with fixed even $h \geq 2$. The class $\mathcal{M}_2(A, B, \epsilon_{\text{NT}})$ consists of nonnegative functions with*

$$\begin{aligned} F(a_1 b_1, a_2 b_2) &\leq \min\{A^{\Omega(a_1 a_2)}, B(a_1 a_2)^{\epsilon_{\text{NT}}}\} F(b_1, b_2), \\ (a_1 a_2, b_1 b_2) &= 1. \end{aligned} \quad (2.10)$$

The functions used here are multiplicative under these coprime products. Put $Q = Q_1 Q_2$, $g = \deg Q = 2$, $\mathfrak{d} = \text{disc}(Q) = h^2$, and let $\|Q\|$ be its maximum coefficient modulus. For fixed $A, B \geq 1$, $0 < \alpha_{\text{NT}}, \delta_{\text{NT}} < 1$ and

$$0 < \epsilon_{\text{NT}} \leq \frac{\alpha_{\text{NT}} \delta_{\text{NT}}}{12g^2}, \quad x \geq c_{\text{NT}} \|Q\|^{\delta_{\text{NT}}}, \quad x^{\alpha_{\text{NT}}} \leq y \leq x, \quad (2.11)$$

the form of the upper bound used is

$$\sum_{x < n \leq x+y} F(n, n+h) \ll_{A, B, \alpha_{\text{NT}}, \delta_{\text{NT}}, g, \mathfrak{d}} y \prod_{p \leq x} \left(1 - \frac{\rho_h(p)}{p} \right) \sum_{ab \leq x} \frac{F(a, b)}{ab}. \quad (2.12)$$

The local root counts of each linear polynomial at every modulus are one. The constants, including the fixed threshold constant c_{NT} , may depend on the indicated parameters and the discriminant; this dependence is harmless for fixed h . They are uniform over F in this common class, hence need not depend on σ or X . Only this upper bound is used.

2.2 Mass and the prime-factor tail

Proof of the mass and tail assertions of Theorem 1.1. Choose a fixed small $\theta > 0$ so that (2.8) with a fixed sufficiently large sieve ratio and $z = X^\theta$ gives $\gg_h X/\log^2 X$ pairs with both endpoints rough. Their factor counts are at most $K_0 = \lceil 1/\theta \rceil + 1$ for large X , and

$$1 - p^{-\sigma} \geq 1 - e^{-c\theta} > 0, \quad W_X(n) \geq (1 - e^{-c\theta})^{2K_0} \quad (2.13)$$

on this support. This proves the lower mass bound without sign information. We obtain this lower bound from the fundamental lemma rather than from a two-sided mean-value theorem: $J_\sigma(2) \sim \sigma \log 2 \rightarrow 0$, so the fixed prime-factor lower-growth hypothesis needed for such a two-sided theorem is unavailable uniformly in X .

For the upper estimates we use the theorem of Nair and Tenenbaum [18] in the form of [14, Theorem 1]. For a marked endpoint set

$$F(a, b) = u^{\Omega(a)} J_\sigma(a) J_\sigma(b), \quad u = 2^{1/1000}. \quad (2.14)$$

This is nonnegative and multiplicative under the coprime products in (2.10). Since $J_\sigma \leq 1$, for $(a_1 a_2, b_1 b_2) = 1$ its growth ratio is

$$\frac{F(a_1 b_1, a_2 b_2)}{F(b_1, b_2)} = u^{\Omega(a_1)} J_\sigma(a_1) J_\sigma(a_2) \leq u^{\Omega(a_1)} \leq a_1^{1/1000} \leq (a_1 a_2)^{1/500}. \quad (2.15)$$

It is also at most $u^{\Omega(a_1 a_2)}$. For the symmetric function $\tilde{F}(a, b) = J_\sigma(a) u^{\Omega(b)} J_\sigma(b)$ the same calculation has a_2 in place of a_1 . Thus both functions belong to $\mathcal{M}_2(u, 1, 1/500)$ with parameters independent of X and σ . The polynomials $n, n + h$ are irreducible and coprime in $\mathbb{Q}[n]$, their product has degree $g = 2$ and fixed discriminant h^2 , and it has no fixed prime divisor: for $p = 2$ odd n avoids both roots, and for $p \geq 3$ there are at most two roots. Take $\alpha_{\text{NT}} = \delta_{\text{NT}} = 1/2$ and $y = x$; then $\alpha_{\text{NT}} \delta_{\text{NT}} / (12g^2) = 1/192$, so $\epsilon_{\text{NT}} = 1/500$ is admissible. For dyadic starting points $x \geq \sqrt{X}$, the threshold $x \geq c_{\text{NT}} \|Q\|^{1/2}$ is satisfied eventually because h is fixed. The 2014 corrigendum [15] concerns the later Theorems 5–6; here we use the Nair–Tenenbaum Theorem 1 with fixed-discriminant constants.

Put $j_p = 1 - p^{-\sigma}$. The positive local prime-power sums are

$$1 + \frac{j_p u}{p - u}, \quad 1 + \frac{j_p}{p - 1}, \quad \frac{1}{\log^2 X} \exp \left((u + 1) \sum_{p \leq X} \frac{1 - p^{-\sigma}}{p} \right). \quad (2.16)$$

The last expression bounds their contribution after the dimension-two sieve product, up to a fixed constant and a bounded h -dependent factor at primes dividing h . The sum in (2.12) is bounded above by the corresponding positive Euler product. The prime-power tails converge uniformly since $u < 2$, including the geometric series $\sum_{\nu \geq 1} (u/2)^\nu$ at the prime 2; the remaining $O(p^{-2})$ terms are summable. Also

$$\sum_{p \leq X} \frac{1 - p^{-\sigma}}{p} \leq \sigma \sum_{p \leq X} \frac{\log p}{p} \ll c. \quad (2.17)$$

Apply Theorem 1 on dyadic intervals with starting point at least $\sqrt{X}$; their logarithms are comparable to $\log X$, so summation gives $O_{h,c}(X/\log^2 X)$. Below $\sqrt{X}$, $J_\sigma \leq 1$ and the growth bound give $O(X^{1/2+1/2000})$. Interchange the marked endpoint to obtain the two terms in (1.2); use $u = 1$ for the unsigned upper bound. Markov's inequality gives (1.3) with the same constant for all K . $\square$

2.3 The two single-sign moments

Proof of (1.4). Fix ε and then choose a large sieve ratio s with a positive error upper bound $0 < E(s) \leq \varepsilon/100$. Set $\eta = 1/4$, $\theta = \eta/s$, $D = X^\eta$, $Y = X^\theta$, and

$$J_{\leq Y}(n) = \prod_{\substack{p \leq Y \\ p|n}} (1 - p^{-\sigma}), \quad U(n) = J_{\leq Y}(n) J_{\leq Y}(n+h). \quad (2.18)$$

At each prime $p \leq Y$, independently for each endpoint, retain its negative divisibility condition with probability $p^{-\sigma}$. The expectation of the resulting two-endpoint sieve indicator is U . For each mark realisation $\mathfrak{M}$, the forbidden roots are a subset of $\{0, -h\}$, so its local densities satisfy $g_{\mathfrak{M}}(p) \leq \rho_h(p)/p$ and $g_{\mathfrak{M}}(2) \leq 1/2$. Consequently the fundamental-lemma constants and the CRT error are independent of the marks; the divisor expansion uses its selected root classes. Since $|\xi_d^\pm| \leq 1$, either signed remainder is at most $2 \sum_{d \leq D} \mu^2(d) \rho_h(d) E_d$, independently of the marks; the factor two allows for the two prefixes when the sign is $\lambda(n+h)$. Let $\bar{V}$ be the average of the local density products; then $\bar{V} \geq V_h(Y) \geq c_*/\log^2(2Y)$. Apply the upper and lower sieve weights to each choice of marks and average. CRT errors are $O(D \log D)$ and main-term errors at most $E(s)X\bar{V}$. For sufficiently large X ,

$$\sum U \geq (1 - 2E(s))X\bar{V}, \quad \sum |\bar{w}^+ - U| \leq 4E(s)X\bar{V}. \quad (2.19)$$

By (2.4), either singly signed averaged upper divisor sum is at most $E(s)X\bar{V}$ in absolute value for large X . Therefore for either single sign g ,

$$\left| \sum g(n)U(n) \right| \leq 5E(s)X\bar{V} \leq 6E(s) \sum U. \quad (2.20)$$

These are positive averaging and L^1 comparisons, so no signed sieve error is assumed.

For $X \geq h$, both endpoints are at most $2X$. Each has at most $\lceil 1/\theta \rceil + 1$ prime factors larger than Y , with multiplicity. Put $B = 2\lceil 1/\theta \rceil + 2$ and choose

$$a = (1 - e^{-c\theta})^B, \quad c_0(\varepsilon) = \theta^{-1} \log(100B/\varepsilon), \quad aU \leq W_X \leq U. \quad (2.21)$$

For fixed $c \geq c_0$, Bernoulli's inequality gives $1 - a \leq \varepsilon/100$. Hence

$$|M_i| \leq (6E(s) + 1 - a) \sum U \leq \frac{6E(s) + 1 - a}{a} M_0 \quad (i = 1, 2). \quad (2.22)$$

Twice the coefficient is less than ε since $2(7\varepsilon/100)/(1 - \varepsilon/100) < \varepsilon$ for $\varepsilon \leq 1/4$. The order is ε , s , θ , c_0 , fixed c, h , then the threshold for X . Thus c_0 is independent of h . $\square$

Remark 2.4 (Reflection). The same marginal conclusion holds for every large even N , with $\sigma = c/\log N$ and $J_\sigma(n)J_\sigma(N-n)$ on $1 \leq n < N$, with a threshold uniform in N . The roots are subsets of $\{0, N\}$, with at most one at 2; the sieve constants and the lower bound by V_2 remain uniform. The substitution $m = N - n$ sends each progression to a progression of the same modulus, so maximal all-residue BV applies to the second sign. Both endpoints are at most N . A bounded- Ω Jordan consumer for varying N would additionally require a uniform tail bound relative to its unsigned mass and the joint contraction. Those additional reflected estimates are not asserted.

Proof of Theorem 1.2. For each (s, t) the exact projection gives

$$\sum_{n \leq X} W_X(n) \mathbf{1}_{(\lambda(n), \lambda(n+h))=(s,t)} = \frac{M_0 + sM_1 + tM_2 + stM_{12}}{4} \geq \frac{\varepsilon}{4} M_0. \quad (2.23)$$

Use (1.2), and only now choose fixed K so that $C'_{h,c}u^{-K} \leq \varepsilon a_{h,c}/8$, for example an integer at least $\log(8C'_{h,c}/(\varepsilon a_{h,c}))/\log u$, enlarged to be nonnegative. Removing the global tail (1.3) leaves at least $\varepsilon a_{h,c}X/(8\log^2 X)$ weighted mass for each pattern. Since $0 < W_X \leq 1$, its unweighted count is at least this large. More generally the same proof works if

$$|M_1| + |M_2| + |M_{12}| \leq (1 - \gamma)M_0, \quad \gamma > 0 \text{ fixed}, \quad (2.24)$$

replacing ε by γ in this projection and tail choice. $\square$

2.4 Consumers at two sieve levels

Proof of Proposition 1.3. Reduce θ to a fixed positive number below $1/4$; positivity preserves the hypothesis. Put $D = X^\theta$. All roots supporting the squarefree upper sieve weight are included in (1.7), and its constant is uniform in all large X . The two upper-weighted single-sign sums are $O_B(X \log^{-B} X)$ by (2.4). Choose a fixed $0 < \delta < \theta$ so small that

$$E(\theta/\delta) < 10^{-4}, \quad 4C_{\text{rem}}\delta^2/c_* < 10^{-3}, \quad z = X^\delta. \quad (2.25)$$

For large X , $\log(2z) \leq 2\delta \log X$, so the upper-weighted joint sign sum is at most $10^{-3}XV_h(z)$. Increase the threshold until $D \log(2D)$ and each of the weighted single-sign remainders are at most $10^{-3}XV_h(z)$. The unsigned rough mass is then at least $0.99XV_h(z)$, and each of the three signed rough sums is at most $0.01XV_h(z)$ by (2.8). The sign identity

$$\mathbf{1}_{\lambda(n)=s, \lambda(n+h)=t} = \frac{1}{4}(1 + s\lambda(n) + t\lambda(n+h) + st\lambda(n)\lambda(n+h)) \quad (2.26)$$

gives at least $XV_h(z)/8$ occurrences on the rough support. There

$$\Omega(n), \Omega(n+h) \leq \frac{\log(2X)}{\delta \log X} \leq 1/\delta + 1, \quad K = \lceil 1/\delta \rceil + 1. \quad (2.27)$$

This and (2.9) prove the assertion, with fixed K even for any finite C_{rem} .

For reflection use $\rho_N(p) = 1$ when $p \mid N$ and 2 otherwise. The factor at 2 is $1/2$, so the sieve constants, CRT error and density lower bound are uniform. The unsigned length is $N - 1$, and the missing endpoint is absorbed in the CRT error. Reflection is a bijection of $\{1, \dots, N - 1\}$ and sends a progression to one of the same modulus. Use the independent hypothesis (1.9) for the joint sign and repeat the fixed- δ argument. A fixed positive-shift bound alone does not supply this moving negative-slope hypothesis.

At the smaller level choose sufficiently large fixed s_0 and

$$D = \exp((\log X)^\alpha), \quad z = \exp((\log X)^\alpha/s_0), \quad (2.28)$$

$$\#\{n \leq X : P^-(n), P^-(n+h) > z, (\lambda(n), \lambda(n+h)) = (s, t)\} \geq c_{h,s_0} \frac{X}{(\log X)^{2\alpha}}.$$

The sieve ratio is s_0 and the relative joint remainder is $O((\log X)^{2\alpha-2}) \rightarrow 0$. BV and CRT errors are negligible at this subpower level. The same projection yields $c_{h,s_0}X/(\log X)^{2\alpha}$ patterns; roughness gives (1.11). For reflection the same calculation uses its separate stretched-exponential root hypothesis and replaces X by N . These bounds grow with the large variable. $\square$

Remark 2.5 (One-point poles under a zero-free input). This remark uses the zero-free half-plane $\Re s > 7/8$ of OpenAI, family 003 [19]; the preceding theorems do not use it. For fixed $c > 0$, $0 < \epsilon < 1/32$, and $\sigma = c/\log X$, one has

$$\begin{aligned} \sum_{n \leq X} J_\sigma(n) &= \frac{cX}{\log X} \{1 + O_c(1/\log X)\} + O_{c,\epsilon}(X^{15/16+\epsilon}), \\ \sum_{n \leq X} \lambda(n) J_\sigma(n) &= -\frac{ce^{-c}X}{\log X} \{1 + O_c(1/\log X)\} + O_{c,\epsilon}(X^{15/16+\epsilon}), \\ \sum_{n \leq X} J_\sigma(n)^2 &= \frac{c(1 - e^{-2c})X}{2\log X} \{1 + O_c(1/\log X)\} + O_{c,\epsilon}(X^{15/16+\epsilon}). \end{aligned} \quad (2.29)$$

The finite identities and Dirichlet series are

$$J_\sigma(n) = \sum_{d|n} \mu(d) d^{-\sigma}, \quad \lambda(n) J_\sigma(n) = \sum_{d|n} \mu^2(d) d^{-\sigma} \lambda(n/d), \quad (2.30)$$

$$\begin{aligned} \sum_n J_\sigma(n) n^{-s} &= \frac{\zeta(s)}{\zeta(s+\sigma)}, \\ \sum_n \lambda(n) J_\sigma(n) n^{-s} &= \frac{\zeta(2s)\zeta(s+\sigma)}{\zeta(s)\zeta(2s+2\sigma)} \quad (\Re s > 1). \end{aligned} \quad (2.31)$$

The second finite identity uses complete multiplicativity and $\mu(d)\lambda(d) = \mu^2(d)$. For large X , $0 < \sigma \leq 1/64$. The logarithm estimate recalled in Remark 6.9 gives arbitrarily small power growth in height for the zeta factors and their reciprocals on the contour shifted to $\Re s = 15/16$. The factors at $2s$ converge absolutely. Perron truncation at height X^3 is uniform because the coefficient moduli are at most one. The crossed poles in the two series are respectively 1 and $1 - \sigma$; the exact summatory residues are

$$\frac{X}{\zeta(1+\sigma)}, \quad \frac{\zeta(2-2\sigma)}{\zeta(1-\sigma)\zeta(2)} \frac{X^{1-\sigma}}{1-\sigma}. \quad (2.32)$$

At $s = 1$ the reciprocal $1/\zeta(s)$ is a zero. Laurent expansion gives $\sigma X(1 + O(\sigma))$ and $-\sigma X^{1-\sigma}(1 + O(\sigma))$. For the square the exact local algebra is

$$\begin{aligned} \sum_n J_\sigma(n)^2 n^{-s} &= \frac{\zeta(s)\zeta(s+2\sigma)}{\zeta(s+\sigma)^2} H(s, \sigma), \\ H(s, \sigma) &= \prod_p \left(1 - \frac{p^{-2\sigma}(1 - p^{-\sigma})^2 p^{-2s}}{(1 - p^{-s-\sigma})^2} \right). \end{aligned} \quad (2.33)$$

This product is analytic and uniformly bounded on $\Re s \geq 15/16$, and $H = 1 + O(\sigma^2)$ there by $1 - p^{-\sigma} \leq \sigma \log p$ and $\sum_p (\log p)^2 p^{-15/8} < \infty$. Its numerator factors are nonzero for $\Re s > 0$. The summatory residues at 1 and $1 - 2\sigma$ are $\sigma X(1 + O(\sigma))/2$ and $-\sigma X^{1-2\sigma}(1 + O(\sigma))/2$. Both poles lie at least $1/32$ to the right of the shifted line; the error is $O_{c,\epsilon}(X^{15/16+\epsilon})$. Since $X^{-\sigma} = e^{-c}$, this proves (2.29). All constants allow fixed c ; no claim for $c = c(X)$ is made. For $F(\beta) = \sum_{n \leq X} \lambda(n) J_\sigma(n) e(n\beta)$, Parseval gives $\int_0^1 |F|^2 \asymp_c X/\log X$, one logarithm larger than the paired mass. The single-site residues are not paired main terms and do not estimate M_{12} .

3 A sequence for the sieve remainder

3.1 Root harmonic mass

For the shift 2, define in addition to (1.6)

$$U_f(X, D) = \sum_{d \leq D} \sum_{r \bmod d} |S_f(X; d, r)|, \quad A_f(X, D) = \sum_{d \leq D} \frac{1}{d} \sum_{r \bmod d} |S_f(X; d, r)|. \quad (3.1)$$

Discarding nonroot and nonsquarefree nonnegative terms, and then using $1 \leq D/d$, gives

$$R_f(X, D) \leq U_f(X, D), \quad R_f(X, D) \leq DA_f(X, D). \quad (3.2)$$

An unnormalised all-residue bound $U_\lambda \ll X \log^{-A} X$, $A > 2$, would supply the needed root bound. The normalised average has a different strength.

Lemma 3.1 (Harmonic root sum). *Let $a(d) = \mu^2(d)|\mathcal{R}_2(d)|$, with value zero on nonsquarefree integers. It is multiplicative and*

$$a(2) = 1, \quad a(p) = 2 \ (p > 2), \quad a(p^k) = 0 \ (k \geq 2). \quad (3.3)$$

For $T \geq 2$,

$$\sum_{d \leq T} a(d) \ll T \log(2T), \quad \sum_{d \leq T} \frac{a(d)}{\sqrt{d}} \ll \sqrt{T} \log(2T), \quad \sum_{d \leq T} \frac{a(d)}{d} \ll \log^2(2T). \quad (3.4)$$

Uniformly for primes $p \geq 5$ and $T \geq 2$,

$$H_p(T) := \sum_{\substack{m \leq T \\ (m, p) = 1}} \frac{a(m)}{m} = \frac{c_2}{2(1 + 2/p)} \log^2 T + O(\log(2T)), \quad (3.5)$$

where c_2 is defined in (1.16). Its implied constant is absolute.

Proof. The roots at an odd prime are $0, -2$; at 2 they coincide. CRT proves multiplicativity. Since $a(d) \leq \tau(d)$,

$$\sum_{d \leq T} a(d) \leq \sum_{u \leq T} \lfloor T/u \rfloor \leq T(1 + \log T). \quad (3.6)$$

Partial summation gives the other two bounds. Define multiplicative b_p by the local polynomials

$$\sum_{k \geq 0} b_p(\ell^k) t^k = \begin{cases} (1 - t)^2, & \ell = p, \\ (1 + t)(1 - t)^2, & \ell = 2, \\ (1 + 2t)(1 - t)^2 = 1 - 3t^2 + 2t^3, & \ell \neq 2, p. \end{cases} \quad (3.7)$$

Coefficient multiplication gives $a(n)\mathbf{1}_{(n, p) = 1} = (\tau * b_p)(n)$, where convolution is over divisors. Uniformly in $p \geq 5$,

$$\sum_{k \geq 1} |b_p(k)| k^{-3/4} \leq C < \infty. \quad (3.8)$$

The generic Euler factors are $1 + 3\ell^{-3/2} + 2\ell^{-9/4}$, with a convergent product; the exceptional factors are absolutely bounded. Consequently $\sum |b_p(k)| \log^j(2k)/k$ is uniformly bounded for $j = 0, 1, 2$. Elementary integral comparison gives, for $y \geq 1$,

$$\sum_{uv \leq y} \frac{1}{uv} = \frac{1}{2} \log^2 y + O(1 + \log y). \quad (3.9)$$

Indeed sum over v , then use $\sum_{u \leq y} 1/u = \log y + O(1)$ and $\sum_{u \leq y} (\log u)/u = \frac{1}{2} \log^2 y + O(1 + \log y)$. Insert this in

$$H_p(T) = \sum_{k \leq T} \frac{b_p(k)}{k} \sum_{uv \leq T/k} \frac{1}{uv}, \quad \sum_k \frac{b_p(k)}{k} = \frac{c_2}{1 + 2/p}. \quad (3.10)$$

Expansion of $(\log T - \log k)^2$ gives the claimed error; the infinite coefficient sum costs $O(T^{-1/4} \log^2 T)$ by (3.8). At 2 its factor is $3/8$ and at the primes $\ell \neq 2, p$ it is $(1 + 2/\ell)(1 - 1/\ell)^2$; at p it is $(1 - 1/p)^2$, the generic factor times $(1 + 2/p)^{-1}$. The logarithm of the positive product converges since the generic factors are $1 + O(\ell^{-2})$. Thus $c_2 > 0$ and $H_p(T) \asymp \log^2 T$ uniformly for sufficiently large T . $\square$

3.2 The independent atoms

Choose an integer j_0 so large that $t_{j_0} = j_0 \log 2 > e^2$ and $\exp(t_{j_0}/\log t_{j_0}) > 5$. For $j \geq j_0$ put

$$\begin{aligned} t_j &= j \log 2, & I_j &= [2^j, 2^{j+1}) \cap \mathbb{Z}, & Y_j &= \exp(t_j/\log t_j), & \kappa_j &= (8 \log t_j)^{-1}, \\ \mathcal{P}_j &= \{p \text{ prime} : Y_j < p \leq 2Y_j\}, & J_j &= |\mathcal{P}_j|. \end{aligned} \quad (3.11)$$

These prime bands overlap; they do not partition the primes. Every sum over j counts a prime once for each band to which it belongs. The independent random atoms below are integer pairs, rather than prime bands. Partition nonnegative integers into the pairs

$$\{4k, 4k+2\}, \quad \{4k+1, 4k+3\} \quad (k \geq 0), \quad \mathcal{M} = \{n \geq 0 : n \equiv 0, 1 \pmod{4}\}. \quad (3.12)$$

At starts $n \in \mathcal{M} \cap I_j$, $j \geq j_0$, prescribe

$$m(n) = \kappa_j \sum_{p \in \mathcal{P}_j} (\mathbf{1}_{p|n} - \mathbf{1}_{p|n-1}); \quad (3.13)$$

at the finitely many earlier starts set $m(n) = 0$. No divisibility of 0 is used here. An integer at most 2^{j+1} has at most $(t_j + \log 2)/\log Y_j \leq 2 \log t_j$ distinct prime factors greater than Y_j . Each of the two nonnegative counts in (3.13) is so bounded, as is their absolute difference. Thus $|m(n)| \leq 1/4$. For each pair choose independently a fair sign Z_n and, independently of Z_n , a sign E_n with

$$\mathbb{P}(E_n = 1) = (1 + m(n))/2, \quad f(n) = Z_n, \quad f(n+2) = Z_n E_n. \quad (3.14)$$

All pairs are independent atoms (Z_n, E_n) . Every nonnegative integer lies in exactly one pair, and pair-block boundaries coincide with dyadic boundaries at least 4, so no prescriptions conflict. Restricting to positive integers gives one infinite function. Its exact moments are

$$\begin{aligned} \mathbb{E}f(u) &= 0, & \mathbb{E}[f(u)f(v)] &= \begin{cases} 1, & u = v, \\ m(\min(u, v)), & \{u, v\} \text{ is a designated pair,} \\ 0, & \text{otherwise,} \end{cases} \\ b(n) &:= \mathbb{E}[f(n)f(n+2)] = \mathbf{1}_{n \in \mathcal{M}} m(n). \end{aligned} \quad (3.15)$$

Within a pair the product is E_n , and every individual sign is unbiased. Different pairs are independent.

3.3 Simultaneous concentration

Lemma 3.2 (Bounded differences). *If a real function Z of finitely many independent atoms changes by at most c_i when atom i changes, then*

$$\mathbb{P}(|Z - \mathbb{E}Z| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_i c_i^2}\right). \quad (3.16)$$

Proof. Reveal the atoms in order and let M_i be the conditional expectation. The i th martingale increment has conditional mean zero and conditional range of length at most c_i . For a mean-zero W in an interval of length c , let $g(s) = \log \mathbb{E}e^{sW}$. The tilted law is in the same interval, so its variance is at most $c^2/4$ by comparison with the interval midpoint. Hence $g''(s) \leq c^2/4$ and $g(0) = g'(0) = 0$ imply $g(s) \leq s^2 c^2/8$. Apply this conditionally and iterate to bound the exponential moment by $\exp(s^2 \sum c_i^2/8)$. Markov with $s = 4t/\sum c_i^2$ gives the upper tail, and $-Z$ gives the lower. If all $c_i = 0$, the function is constant and the claim is immediate. $\square$

Lemma 3.3 (One realisation for all queries). *Almost surely (1.12) and (1.13) hold eventually, and for every pair of integer affine maps with nonzero slopes and coefficients of modulus at most M , and every interval J of at most M parameters with arguments in $[1, M]$,*

$$\left| \sum_{n \in J} \{f(L_1(n))f(L_2(n)) - \mathbb{E}[f(L_1(n))f(L_2(n))]\} \right| \leq C\sqrt{|J|\log(2M)}. \quad (3.17)$$

The same holds when the parameter interval is intersected with a progression, replacing $|J|$ by its actual number of parameters. Empty sums are zero.

Proof. For a real or imaginary part of a one-point sum with t terms, an atom changes it by at most 4, and at most t atoms occur. The square-influence sum is at most $16t$, so its tail is $2 \exp(-u^2/(8t))$. In a two-point test each vertex occurs at most once in each affine map. An atom has two vertices, affects at most four terms and changes each by at most 2. At most $2|J|$ atoms occur, giving square-influence sum at most $128|J|$ and tail $2 \exp(-u^2/(64|J|))$. Progression restrictions preserve injectivity. Coincident arguments give deterministic terms and do not increase this bound.

Intervals and progressions in $[1, M]$ have $O(M^4)$ choices. For the two-point catalogue separate zero, one and at least two terms. A singleton is specified by two values in $[1, M]$, giving at most M^2 tests; its centred sum has modulus at most 2. For $t \geq 2$ the actual output lists have the form

$$x_k = x_0 + k\Delta_1, \quad y_k = y_0 + k\Delta_2 \quad (0 \leq k < t), \quad M^2(2M-1)^2M = O(M^5) \quad (3.18)$$

possible lists: the initial values lie in $[1, M]$, $0 < |\Delta_i| \leq M-1$ by the value range, and $t \leq M$. This counts the tests rather than their possibly many parameter descriptions. It leaves the influence budget unchanged and is smaller than the crude $O(M^{10})$ bound.

For additive frequencies use k/M^3 , $0 \leq k < M^3$. Nearest-grid interpolation costs at most $2\pi tM/M^3 \leq 2\pi/M$, so the one-point catalogue has $O(M^7)$ tests. Use thresholds $200\sqrt{t\log(2M)}$ and $200\sqrt{|J|\log(2M)}$. The corresponding single-query real tails are bounded by $2(2M)^{-5000}$ and $2(2M)^{-625}$; splitting a complex sum into its real and imaginary parts only changes constants. The catalogue union probabilities are summable over every integer M . Borel–Cantelli gives all the eventual statements, without independence between queries or scales. For $q > M$ a progression contains at most one term and is covered trivially.

For fixed integer $B \geq 1$, the character catalogue has $O(M^{2B})$ characters, M^2 intervals and $O(M^{B+3})$ height points at spacing M^{-3} in $|v| \leq M^B$: in all $O(M^{3B+5})$ tests. The weights $\chi(n)n^{iv}$

have modulus at most one, so the same $16|I|$ influence budget applies. Moving the height to a nearest point costs at most $|I|M^{-3} \log M \leq M^{-2} \log M$. Threshold $200(B+1)\sqrt{|I| \log(2M)}$ gives a summable union probability. Take the countable probability-one intersection over integer B ; upper integer rounding includes all fixed real $B > 0$. Thus one infinite realisation works for all the stated query catalogues. $\square$

Proposition 3.4 (Affine and reflected means). *The fixed affine, polylogarithmic coefficient, all-shift and reflected estimates of Theorem 1.4 hold for every realisation in Lemma 3.3. In particular, for fixed positive-slope nonproportional forms,*

$$\sum_{n \leq X} f(a_1 n + b_1) f(a_2 n + b_2) \ll_{a_1, a_2, b_1, b_2, f} \sqrt{X \log(2X)}. \quad (3.19)$$

Proof. For different slopes the equations $L_1 - L_2 = 0, 2, -2$ have at most three solutions in total, so the mean is at most 3. For equal slopes different intercepts give zero mean unless they differ by ± 2 . In this remaining case write the smaller argument as $an + b$. On I_j , membership in $\mathcal{M}$ splits the parameters into at most two classes modulo $4/(a, 4)$. If $p \in \mathcal{P}_j$ and $p \nmid a$, CRT gives a difference at most one between the counts $p \mid an + b$ and $p \mid an + b - 1$ in each class. The contribution is $O(\kappa_j J_j)$. Bands containing a prime divisor of fixed a lie in a finite initial range and cost $O_a(1)$. For large M , monotonicity of $t/\log t$ and $J_j \leq 2Y_j$ give

$$Z(M) := 1 + \sum_{j_0 \leq j \leq \log_2 M} \kappa_j J_j \leq \exp(2 \log M / \log \log M) = M^{o(1)}. \quad (3.20)$$

With M a fixed multiple of $X + 1$, the mean is $O_{a,b}(1 + Z(M)) = o(\sqrt{X})$. Lemma 3.3 proves (3.19); a fixed progression changes only the fixed coefficients and starting index. For reflection, only $2n - N = 0, \pm 2$ has nonzero mean, at at most three positions. The simultaneous query event with ambient $M = N$ gives (1.14) for all totals.

For slopes bounded by $(\log X)^B$, a prime divisor of the common slope is at most that size. A band containing such a prime satisfies $t_j / \log t_j \leq B \log \log X$, hence $t_j = O_B(\log \log X \log \log \log X)$ and its whole support is $X^{o_B(1)}$. Other bands cost at most $Z(2X(\log X)^B) = X^{o_B(1)}$; the deviation remains $O(\sqrt{X \log X})$. This proves the uniform coefficient version without a fixed-form threshold hidden in growing coefficients. For $1 \leq h \leq X$, the mean is zero unless $h = 2$; the latter has just been estimated. Use ambient $2X$ in the same event to obtain the all-shift bound. $\square$

3.4 The two residue averages

Proposition 3.5 (Root amplification). *Every realisation in Lemma 3.3 satisfies (1.15), (1.16) and (1.17) for every fixed parameter in their stated ranges.*

Proof. Write $L_j = |I_j \cap [1, X]|$. For squarefree d and every $r \bmod d$, CRT gives

$$B(X; d, r) := \sum_{\substack{n \leq X \\ n \equiv r \pmod{d}}} b(n) = \frac{1}{2} \sum_j \frac{\kappa_j L_j}{d} \sum_{\substack{p \in \mathcal{P}_j \\ p \mid d}} (\mathbf{1}_{r \equiv 0 \pmod{p}} - \mathbf{1}_{r \equiv 1 \pmod{p}}) + O(Z(X)). \quad (3.21)$$

Only nonempty intervals $I_j \cap [1, X]$ are summed. For odd d the progression meets four classes modulo 4, two allowed; for even squarefree d it meets two, one allowed. This explains the factor $1/2$. If $p \nmid d$, the two divisibility tests have the same density and differ by $O(1)$ per interval; if $p \mid d$, their indicators are fixed and the matching-start count is $L_j/(2d) + O(1)$. The error is uniform even when $dp > X$.

At a root and $p \geq 5$ dividing d , the residue is 0 or -2 , never 1. Every main term is nonnegative and exactly half the roots have residue 0 at p . Put $d = pm$; then $\rho_2(pm) = 2\rho_2(m)$ for squarefree $(m, p) = 1$. The summed main quantity is

$$\mathcal{B}(X, D) = \frac{1}{2} \sum_j \kappa_j L_j \sum_{\substack{p \in \mathcal{P}_j \\ p \leq D}} \frac{1}{p} H_p(D/p). \quad (3.22)$$

Taking absolute values costs only

$$O(DZ(X) \log(2D)), \quad (3.23)$$

by (3.4) and $||v + e| - v| \leq |e|$ for $v \geq 0$. The prime number theorem [17], by partial summation on dyadic prime intervals, gives

$$J_j \asymp Y_j / \log Y_j, \quad \sum_{p \in \mathcal{P}_j} p^{-1} \asymp 1 / \log Y_j, \quad \sum_{p \in \mathcal{P}_j} p^{-2} \ll 1 / (Y_j \log Y_j). \quad (3.24)$$

For $D = X^\theta$, a full dyadic interval $I_j \subset [1, X]$ with $X/4 < 2^j \leq X/2$ has $L_j \asymp X$ and

$$\log p \sim \frac{\log X}{\log \log X} = o(\log X), \quad \log(D/p) = (\theta + o(1)) \log X, \quad \kappa_j \sum_{p \in \mathcal{P}_j} p^{-1} \asymp 1/t_j \asymp 1/\log X. \quad (3.25)$$

Thus (3.5) gives $\mathcal{B} \gg_\theta X \log X$. For the upper bound $H_p(D/p) \ll \log^2(2D)$ and

$$\sum_{j \leq \log_2 X} \kappa_j L_j \sum_{p \in \mathcal{P}_j} p^{-1} \ll X / \log(2X). \quad (3.26)$$

Split off $j \leq \frac{1}{2} \log_2 X$, with total length $O(\sqrt{X})$; on later bands $\kappa_j / \log Y_j = 1/(8t_j) \ll 1/\log X$. Small fixed bands are absorbed. Therefore $\mathcal{B} \asymp_\theta X \log X$.

The progression-restricted concentration estimate gives, simultaneously for every $d \leq D$ and r ,

$$|S_f(X; d, r) - B(X; d, r)| \ll \sqrt{(X/d + 1) \log(2X)}. \quad (3.27)$$

Summing over roots yields

$$O\left(\sqrt{XD} \log^{3/2}(2X) + D \log^{3/2}(2X)\right). \quad (3.28)$$

For fixed $\theta < 1$ both errors are $o(X \log X)$, proving $R_f(X, X^\theta) \asymp_\theta X \log X$.

For a general d let $c(d, r) \in \{0, 1/2, 1\}$ be the density of starts in the progression: it is $1/2$ if $4 \nmid d$ and the relevant 0 or 1 if $4 \mid d$. The analogue of (3.21) uses $c(d, r)$ instead of $1/2$. For $p \mid d$,

$$\begin{aligned} \sum_{r \bmod d} |\mathbf{1}_{r \equiv 0 \pmod{p}} - \mathbf{1}_{r \equiv 1 \pmod{p}}| &= 2d/p, \\ \sum_{d \leq D} \frac{1}{d} \sum_r |B(X; d, r)| &\ll \log(2D) \sum_j \kappa_j L_j \sum_{p \in \mathcal{P}_j} p^{-2} + DZ(X). \end{aligned} \quad (3.29)$$

Here $\sum_{d \leq D, p \mid d} 1/d \leq (1 + \log(2D))/p$. Let $t = \log X$. Bands with $t_j \leq t/2$ cost $O(\sqrt{X} \log(2X))$. On the others,

$$\kappa_j \sum_{p \in \mathcal{P}_j} p^{-2} \ll \frac{1}{tY_j} \ll \frac{1}{t} \exp\left(-\frac{t}{2 \log t}\right). \quad (3.30)$$

Their total, using $\log D \leq t$ and $\sum L_j \leq X$, is at most $CX \exp(-t/(2 \log t))$. The normalised fluctuation costs

$$\sum_{d \leq D} \sqrt{(X/d + 1) \log(2X)} \ll \sqrt{XD \log(2X)} + D \sqrt{\log(2X)}. \quad (3.31)$$

At $D = X^\theta$, $\theta < 1$ fixed, this and $DZ(X)$ are $O_\theta(X^{1-\delta_\theta})$ for some $\delta_\theta > 0$. They are smaller than $X \exp(-t/(4 \log t))$; this proves (1.15), with all moduli included.

For the exact constant retain only $t_j \geq t - \sqrt{t}$. Earlier bands have total length $O(Xe^{-\sqrt{t}})$ and cost $O(Xe^{-\sqrt{t}}t^2) = o(Xt)$ in (3.22). On the retained bands, uniformly in j, p ,

$$\begin{aligned} t_j &= t + O(\sqrt{t}), \quad \log(D/p) = (\theta + o(1))t, \quad H_p(D/p) = (c_2\theta^2/2 + o(1))t^2, \\ \sum_{Y_j < p \leq 2Y_j} \frac{1}{p} &= \frac{\log 2 + o(1)}{\log Y_j}, \quad \kappa_j \sum_{p \in \mathcal{P}_j} \frac{1}{p} = \frac{\log 2 + o(1)}{8t}. \end{aligned} \quad (3.32)$$

The retained lengths sum to $X + o(X)$. The outer $1/2$ in (3.22), the harmonic $1/2$, the $1/8$ in κ_j , the prime-band $\log 2$ and the factor θ^2 from $\log^2(D/p)$ give $c_2 \log 2 \theta^2 X t / 32 + o(Xt)$. Both errors remain $o(Xt)$. These are deterministic estimates on one simultaneous concentration event, so the exact asymptotic holds for every fixed real θ without an uncountable intersection.

For $2 \leq D \leq \exp((\log X)^\alpha)$, fixed $\alpha < 1$, a contributing band must have $Y_j < D$. Put $u = \log D$. Monotonicity of $t/\log t$ gives, for large u ,

$$t_j < 2u \log(2u), \quad \mathcal{B}(X, D) \ll \exp(Cu \log(2u)) \log^2(2D) = X^{o_\alpha(1)}. \quad (3.33)$$

Bounded u needs only a fixed additive constant. The total contributing length is at most $C \exp(2u \log(2u))$. The deterministic error is $X^{o_\alpha(1)}$ and the random error is $X^{1/2+o_\alpha(1)}$, uniformly in the stated range, proving (1.17). Integer floors affect only endpoints already included in these bounds. $\square$

Remark 3.6 (The strength of the normalised saving). The model does not have a fixed power saving for A_f . Fix $0 < \theta < 1$ and take a complete dyadic interval I_j with $X/4 < 2^j \leq X/2$. For large X a prime $p \in \mathcal{P}_j$ exists, and $p \leq 2Y_j < X^\theta$. All root main terms are nonnegative, so $B(X; p, 0) \geq \kappa_j L_j / (2p) - CZ(X)$. By (3.27), its two errors are $o(\kappa_j L_j / p)$. Consequently

$$\begin{aligned} A_f(X, X^\theta) &\geq \frac{|S_f(X; p, 0)|}{p} \geq \frac{\kappa_j L_j}{4p^2} \geq \frac{X}{512 Y_j^2 \log t_j} \\ &= \frac{X}{512 \log \log X} \exp\left(-(2 + o(1)) \frac{\log X}{\log \log X}\right). \end{aligned} \quad (3.34)$$

Thus $A_f/X^{1-\delta} \rightarrow \infty$ for every fixed $\delta > 0$. The upper bound (1.15) saves beyond every fixed inverse power of $\log X$. It does not simulate the stronger premise $A_\lambda \leq X^{1-\delta}$. Conversely, suppose a sequence satisfies $A_f(X, X^{\theta_0}) \ll X^{1-\delta_0}$ for fixed $\theta_0, \delta_0 > 0$. Monotonicity of A_f in its second argument and (3.2) give

$$\theta_1 = \min(\theta_0, \delta_0/2) > 0, \quad R_f(X, X^{\theta_1}) \leq X^{\theta_1} A_f(X, X^{\theta_0}) \ll X^{1-\delta_0/2} \ll X / \log^2 X. \quad (3.35)$$

Thus an actual fixed power saving for the normalised average does provide a positive-level root bound; the model concerns the weaker displayed saving.

Corollary 3.7 (Further numerical bounds). *The same sequence satisfies, for $X \geq H \geq q \geq 1$,*

$$\int_X^{2X} \sum_{r \bmod q} \left| \sum_{\substack{x < n \leq x+H \\ n \equiv r \pmod{q}}} f(n) \right|^2 dx \ll X(H+q) \log(2X) \ll XH \log(2X). \quad (3.36)$$

If $H/q \geq \exp((\log X)^\nu)$ for fixed $\nu > 0$, this is smaller eventually than

$$\frac{XH^2}{q} \{\log(H/q)\}^{-1/1000}. \quad (3.37)$$

For fixed $0 < \theta < 1$, uniformly over integer $1 \leq h \leq X$, $h \neq 2$,

$$R_f^{(h)}(X, X^\theta) \ll_f X^{(1+\theta)/2} \log^{3/2}(2X). \quad (3.38)$$

Proof. Square (1.12), sum progression lengths and integrate using ambient $3X$. The comparison follows from $\log(2X)\{\log(H/q)\}^{1/1000} = o(H/q)$. For $h \neq 2$ the two vertices belong to different components, so every mean is zero. Apply the uniform two-point deviation on ambient $2X$ and sum the roots, bounded by $\tau(d)$. This is (3.28); its D term is smaller than its $\sqrt{XD}$ term for $D \leq X$. $\square$

Corollary 3.8 (Scope of the numerical premises). *The numerical properties listed in the introduction, even together with Corollary 3.7, do not imply a root bound $C_{\text{rem}}X/\log^2 X$ at any fixed positive power level for arbitrary sign sequences with those properties.*

Proof. Fix one realisation in the probability-one event above. It satisfies all these premises, but (1.16) exceeds that bound for every fixed $0 < \theta < 1$ and finite C_{rem} . Levels $\theta \geq 1$ fail by monotonicity from $\theta = 1/2$. Any chain of identities and inequalities valid for arbitrary bounded sign sequences also applies to this realisation. The conclusion concerns only the displayed strengths of the premises. $\square$

Remark 3.9 (Multiplicativity and small levels). Almost surely, infinitely many distinct squares k^2 , $k \geq 2$, have each sign: successive squares are separated by at least 5, so their matching components are different and their signs are independent and fair. The probability of a constant tail is zero, and the countable union over starting indices remains null. A completely multiplicative sign function has $f(k^2) = 1$. The same argument applies to 3^{2k} and a sequence of primes separated by more than 2; hence $f(3^{2k}) = 1$ for all k and $f(p) = -1$ for all primes p both fail. These probability-one events may be intersected with all the preceding events.

More explicitly, for fixed $a \geq 2$, even $h \geq 2$ and a fixed cutoff z with $(a, \prod_{p \leq z} p) = 1$, CRT provides a progression of n with $P^-(n), P^-(n+h) > z$. Choose a rapidly increasing subsequence so that the matching components of $\{n, n+h, an, a(n+h)\}$ are disjoint across choices. For large n the upper two arguments are in different components since their distance is $ah \geq 4$, and are independent fair signs independent of the lower product. Hence

$$f(an)f(a(n+h)) = f(n)f(n+h) \quad (3.39)$$

fails with probability $1/2$ independently at the selected n , and fails infinitely often almost surely. A countable intersection over a, h and finite prime sets preserves one realisation.

For fixed $0 < \rho < \alpha < 1$, put $z = \exp((\log X)^\rho)$ and $D = \exp((\log X)^\alpha)$. The one-point root-weighted error from (1.12) is $X^{1/2+o(1)}$, and (1.17) gives the joint root error at shift 2 of the

same size. The sieve ratio $(\log X)^{\alpha-\rho}$ tends to infinity; the CRT error $D \log D = X^{o(1)}$ and the upper-minus-rough error are $o(XV_2(z))$. The projection therefore gives

$$\#\{n \leq X : P^-(n), P^-(n+2) > z, (f(n), f(n+2)) = (s, t)\} = (1/4 + o(1))XV_2(z) \quad (3.40)$$

for each fixed pattern. The roughness bound on Ω still grows. This is compatible with the smaller-level numerical conclusions; f is not $(-1)^\Omega$.

4 A sequence for Type II sums

4.1 Operator and fourth-moment norms

For $K \in \mathbb{C}^{r \times s}$ define

$$E(K) = \sum_{a,b=1}^s \left| \sum_{m=1}^r \overline{K_{ma}} K_{mb} \right|^2 = \text{tr}((K^* K)^2) = \sum_j \sigma_j^4, \quad (4.1)$$

where σ_j are singular values. Here $\|K\|_F$ denotes the Frobenius norm.

Lemma 4.1 (Norm comparisons). *Every complex matrix satisfies*

$$\|K\|_{\text{op}}^4 \leq E(K) \leq \|K\|_{\text{op}}^2 \|K\|_F^2. \quad (4.2)$$

For a nonempty matrix whose entries have modulus one,

$$\|K\|_{\text{op}} \geq \sqrt{\max(r, s)}, \quad E(K) \geq \max(r^2 s, r s^2). \quad (4.3)$$

Consequently, for $L > 1$, $E(K) \leq (rs)^2 L^{-6-\eta}$ gives $\|K\|_{\text{op}} \leq \sqrt{rs} L^{-3/2-\eta/4}$, whereas the operator bound $\sqrt{rs} L^{-3/2-\kappa}$ gives only $E(K) \leq (rs)^2 L^{-3-2\kappa}$ directly. With constants one these targets require respectively $\min(r, s) \geq L^{6+\eta}$ and $\min(r, s) \geq L^{3+2\kappa}$.

Proof. The positive semidefinite matrix $K^* K$ has eigenvalues σ_j^2 , proving (4.1). Its largest fourth power is at most their sum, and each fourth power is at most $\|K\|_{\text{op}}^2 \sigma_j^2$. A standard column has norm $\sqrt{r}$, and a standard column of K^* has norm $\sqrt{s}$. The diagonal terms of E give rs^2 ; use $\text{tr}((K^* K)^2) = \text{tr}(K K^* K K^*)$ to get rs^2 . Substitution and fourth roots give the consequences. The general comparison also holds for empty matrices. $\square$

Remark 4.2. For $H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $H^* H = 2I$, so $\|H\|_{\text{op}}^4 = 4$ and $E(H) = 8$. Thus an operator estimate and a fourth-moment estimate are not equal with the same exponent and constant in general.

4.2 An infinite construction

Proof of Theorem 1.5. Fix $h \neq 0$ and choose an integer $T_0 \geq \max(16, 2|h| + 16)$, with $T_{j+1} = T_j^2$. Let $\mathcal{P}_j$ be the primes in $(T_j, 2T_j]$ and $\mathcal{Q}_j$ those in $(3T_j, 6T_j]$; define

$$\mathcal{C}_j = \{pq + h : p \in \mathcal{P}_j, q \in \mathcal{Q}_j\} \subset (3T_j^2 + h, 12T_j^2 + h]. \quad (4.4)$$

All arguments are positive. Successive numerical supports are disjoint, as are the two prime catalogues both within and across scales. A value $pq + h$ uniquely determines p, q by unique factorisation and the separated prime intervals, which exclude an exchange of positions. For every (j, p) and

(j, q) take mutually independent fair signs $u_{j,p}, v_{j,q}$; for each positive integer $t \notin \bigcup_j \mathcal{C}_j$ take an additional independent fair sign ϵ_t . Define for every positive integer

$$f(pq + h) = u_{j,p}v_{j,q} \quad (pq + h \in \mathcal{C}_j), \quad f(t) = \epsilon_t \quad (t \notin \bigcup_j \mathcal{C}_j). \quad (4.5)$$

The uniqueness and disjointness just proved make this a single well-defined function. Each value has mean zero and distinct values have zero two-point mean. In a common rectangle two distinct edges share at most one endpoint; an unshared fair sign makes their product mean zero. Fourth-order products are not independent: $f(pq + h)f(pq' + h)f(p'q + h)f(p'q' + h) = 1$ for $p, p' \in \mathcal{P}_j$, $q, q' \in \mathcal{Q}_j$.

For any affine test in (1.18), injectivity of each form makes each value occur at most twice. Let $k_t \in \{0, 1, 2\}$ be the number of terms containing the unplanted atom ϵ_t . Changing it has influence $c_t \leq 2k_t$, and $\sum_t k_t \leq 2|I|$. Since $k_t^2 \leq 2k_t$,

$$\sum_t c_t^2 \leq 4 \sum_t k_t^2 \leq 8 \sum_t k_t \leq 16|I| \leq 16Y. \quad (4.6)$$

At scale j , each $u_{j,p}$ affects at most $3T_j$ values, with influence at most $12T_j$, and there are at most T_j such atoms. Each $v_{j,q}$ has influence at most $4T_j$, with at most $3T_j$ atoms. The scale contributes at most $144T_j^3 + 48T_j^3 = 192T_j^3$. Only scales with $\mathcal{C}_j \cap [1, Y] \neq \emptyset$ contribute. For $Y \geq |h|$ these have $3T_j^2 < Y + |h| \leq 2Y$, so $T_j < \sqrt{Y}$, also when $h < 0$. Since $T_{j+1} \geq 16T_j$,

$$\sum_{j: \mathcal{C}_j \cap [1, Y] \neq \emptyset} T_j^3 \leq 2Y^{3/2}, \quad \sum_i c_i^2 \leq 16Y + 384Y^{3/2} \leq 400Y^{3/2}. \quad (4.7)$$

One-point means vanish. A two-point mean can be nonzero only where $an + b = cn + d$; distinct forms have at most one such parameter, so its modulus is at most 1.

At integer Y there are at most $1000Y^6$ tests, including coefficients, intervals, positive and negative slopes and the value-range conditions. With deviation $\vartheta = 100Y^{3/4}\sqrt{\log(2Y)}$, Lemma 3.2 gives each failure probability at most $2(2Y)^{-50}$ and union probability at most

$$2000Y^6(2Y)^{-50}, \quad \sum_{Y \geq Y_h} 2000Y^6(2Y)^{-50} < 1 \quad (4.8)$$

for large enough Y_h . The countable union bound therefore leaves positive probability for an event where every test at every integer $Y \geq Y_h$ passes. On this event one may take $C_h = 100$; finite initial ranges can be absorbed by enlarging the constant. This chooses a single infinite function.

We can include the character means on the same probability space. For each fixed integer $A \geq 1$ and integer Y , consider real and imaginary parts of $\sum_{n \leq z} f(n)\chi(n)$, for all $q \leq Y^A$, $\chi \bmod q$ and $z \leq Y$. There are at most $2Y^{2A+1}$ tests, since $\varphi(q) \leq q$. Their means are zero and their squared influence is still at most $400Y^{3/2}$. Use threshold

$$\vartheta_A = 100(A+1)Y^{3/4}\sqrt{\log(2Y)}, \quad \mathbb{P}(\text{a character test fails at } Y) \leq 4Y^{2A+1}(2Y)^{-50(A+1)^2}. \quad (4.9)$$

For each A the sum over Y converges. The probability of infinitely many failures is at most every such series tail and hence zero. Intersect these probability-one events over positive integers A , and then intersect with the positive-probability affine event; the resulting event still has positive probability. Fix a realisation in it. Real and imaginary bounds control the modulus, finite initial ranges are absorbed in the constant, and $Y = \lceil x \rceil$, $z = \lfloor x \rfloor$ and upper integer rounding of A give

$$\sum_{n \leq x} f(n)\chi(n) \ll_{h,A} x^{3/4}\sqrt{\log(2x)}, \quad q \leq x^A. \quad (4.10)$$

This is a bound for one-point sums only.

Put

$$A_j = (T_j, 2T_j] \cap \mathbb{Z}, \quad B_j = (3T_j, 6T_j] \cap \mathbb{Z}, \quad K_j(m, \ell) = f(m\ell + h). \quad (4.11)$$

On the prime subrectangle $\mathcal{P}_j \times \mathcal{Q}_j$, the matrix is exactly $uv^\top$. Test with $\alpha_p = u_{j,p}/\sqrt{|\mathcal{P}_j|}$ and $\beta_q = v_{j,q}/\sqrt{|\mathcal{Q}_j|}$, supported on these coordinates, to obtain

$$\left| \sum_{m, \ell} \alpha_m \beta_\ell f(m\ell + h) \right| = \sqrt{|\mathcal{P}_j| |\mathcal{Q}_j|}, \quad |\mathcal{P}_j| \sim T_j / \log T_j, \quad |\mathcal{Q}_j| \sim 3T_j / \log T_j. \quad (4.12)$$

The prime number theorem gives the last two comparisons and proves (1.20). Here $M = |A_j|$ and $N = |B_j|$, the products $m\ell$ have size $X \asymp T_j^2$, and $MN \asymp X$. The ratio of the lower bound to the target $\sqrt{MN}(\log X)^{-3/2-\kappa}$, for any fixed $\kappa > 0$, tends to infinity like at least a positive multiple of $(\log T_j)^{1/2+\kappa}$. By (4.2), the whole matrix also has

$$E(K_j) \gg_h T_j^4 / (\log T_j)^4. \quad (4.13)$$

This exceeds the fourth-moment target $(|A_j| |B_j|)^2 (\log X)^{-6-\kappa}$ by a factor at least a positive multiple of $(\log T_j)^{2+\kappa}$.

For (1.19), $qn + r \leq q(L+1) \leq 2L^{1+\tau}$. Use $Y = \lceil 2L^{1+\tau} \rceil$ to get $O_{h,\tau}(L^{3(1+\tau)/4} \sqrt{\log(2L)})$. The exponent difference is

$$\frac{15+7\tau}{16} - \frac{3+3\tau}{4} = \frac{3-5\tau}{16} > 0 \quad (0 \leq \tau < 1/7), \quad (4.14)$$

so the logarithm is absorbed uniformly in q, r . Fixed affine pairs use ambient Y a fixed multiple of L and give $O_{h,a,b,c,d}(L^{3/4} \sqrt{\log(2L)})$. Reflection uses $(a, b, c, d) = (1, 0, -1, N)$, $Y = N$ and $I = [1, N-1]$, giving $O_h(N^{3/4} \sqrt{\log(2N)})$ for every sufficiently large N .

Finally choose $T_0 \geq 20$ and use $pq+1$ on even scales and $pq+2$ on odd scales. Their numerical supports remain disjoint, each within-scale prescription is unique, and distinct values remain pairwise orthogonal. Since $|h_j| \leq 2$, the same influence budget, test count and positive-probability event apply. Each shift has infinitely many of its own scales, and the character probability-one intersection is unchanged. Thus one f has both asserted families of lower bounds, on alternating scales. $\square$

Remark 4.3 (A dyadic product window). The original rectangles have products in $(3T_j^2, 12T_j^2]$. The same realisation also has the operator lower bound on

$$A'_j = (T_j, 11T_j/10] \cap \mathbb{Z}, \quad B'_j = (3T_j, 33T_j/10] \cap \mathbb{Z}, \quad X_j = 3T_j^2. \quad (4.15)$$

Their products lie in $(3T_j^2, 363T_j^2/100] \subset (X_j, 2X_j]$. The prime number theorem gives

$$|\mathcal{P}_j \cap A'_j| \sim \frac{T_j}{10 \log T_j}, \quad |\mathcal{Q}_j \cap B'_j| \sim \frac{3T_j}{10 \log T_j}. \quad (4.16)$$

The planted submatrix on these prime sets is still $uv^\top$. The same supported unit-vector test as in (4.12) therefore yields, with a different positive constant,

$$\|(f(m\ell + h))_{A'_j \times B'_j}\|_{\text{op}} \geq \sqrt{|\mathcal{P}_j \cap A'_j| |\mathcal{Q}_j \cap B'_j|} \gg_h \frac{T_j}{\log T_j} \asymp \frac{\sqrt{|A'_j| |B'_j|}}{\log X_j}. \quad (4.17)$$

This restriction also applies separately on each of the alternating $h = 1, 2$ scales and requires no additional probabilistic event.

Proposition 4.4 (Additional bounds for the same sequence). *For fixed $0 \leq \theta < 1/3$, uniformly in $1 \leq a \neq b \leq L^\theta$,*

$$\sum_{m \leq L} f(am + h)f(bm + h) \ll_{h,\theta} L^{1-(1-3\theta)/4} \sqrt{\log(2L)}. \quad (4.18)$$

For fixed $0 < \epsilon < 1/4$ and integer $Y \geq Y_h$,

$$\sum_{q \leq Y^{1/4-\epsilon}} \max_{r \bmod q} \max_{z \leq Y} \left| \sum_{\substack{1 \leq n \leq z \\ n \equiv r \pmod{q}}} f(n) \right| \ll_h Y^{1-\epsilon} \sqrt{\log(2Y)}. \quad (4.19)$$

The latter holds for fixed translates too, after omitting nonpositive arguments.

Proof. For (4.18), take $Y = \lceil 2L^{1+\theta} + 2|h| + 2 \rceil$ in the two-point affine bound. Its right side is $O_h(L^{3(1+\theta)/4} \sqrt{\log(2L)})$ and the possible one-parameter collision is absorbed. When $h < 0$ only a fixed $O_h(1)$ initial segment has nonpositive arguments.

For (4.19) choose $1 \leq r \leq q$ and write the progression as $qk + (r - q)$ for $1 \leq k \leq \lfloor (z - r)/q \rfloor + 1$, with empty intervals interpreted as zero. The coefficients and range satisfy the affine hypotheses, so each maximum is at most $C_h Y^{3/4} \sqrt{\log(2Y)}$. Multiply by at most $Y^{1/4-\epsilon}$ moduli. For fixed translates replace ambient Y by $Y + 2|h| + 2$ and delete a fixed initial segment; its cumulative cost $O_h(Y^{1/4-\epsilon})$ is absorbed. In particular $\epsilon = 1/20$ gives level $Y^{1/5}$ and error $O_h(Y^{19/20} \sqrt{\log Y})$. $\square$

Remark 4.5 (The prime subblock and multiplicativity). For any matrix supported on $P \times Q$ with entries of modulus at most one,

$$\|A\|_{\text{op}} \leq \|A\|_{\text{F}} \leq \sqrt{|P||Q|}. \quad (4.20)$$

For a unit vector, rowwise Cauchy–Schwarz proves the bound; $A_{pq} = u_p v_q$ attains equality. Thus the prime subblock above has exactly this norm and relative scale $1/\log X$. Boundedness alone on that subblock cannot improve its cost to $o(\sqrt{MN}/\log X)$. This does not give an optimal global operator upper bound for every sequence satisfying the affine estimates.

For every fixed integer $d \geq 2$, use the distinct forms dn, n , ambient $\lceil dT \rceil$, to get

$$\begin{aligned} \sum_{n \leq T} f(dn)f(n) &= O_d(T^{3/4} \sqrt{\log(2T)}), \\ \#\{n \leq T : f(dn) = sf(n)\} &= \tfrac{1}{2}T + O_d(T^{3/4} \sqrt{\log(2T)}) \quad (s \in \{\pm 1\}). \end{aligned} \quad (4.21)$$

The second formula follows from the indicator $(1 + sf(dn)f(n))/2$. At every fixed prime the relation $f(pn) = f(p)f(n)$ thus fails on asymptotically half the parameters; the Liouville relation $\lambda(pn) = -\lambda(n)$ would hold at every parameter. For a single shift h , every $p + h$ with p prime lies outside the prescribed sets, since $p + h = rs + h$ would force $p = rs$. These positions carry independent ϵ signs, and both signs occur infinitely often almost surely. In the alternating $h = 1, 2$ construction cross-shift cores are excluded by parity ($rs \pm 1$ is even and larger than 2), so the same countable probability-one event can be included without changing the lower bounds.

Corollary 4.6 (Scope of the Type II premises). *The numerical properties in Theorem 1.5 and Proposition 4.4, together with finite algebra, norm inequalities, prime counting and sieve operations on the indices, do not imply for arbitrary sign sequences the operator target $\sqrt{MN}(\log X)^{-3/2-\kappa}$ for fixed $\kappa > 0$.*

Proof. The single sequence just constructed satisfies these premises and $K(m, \ell) = f(m\ell + h)$, but violates the target on infinitely many specified rectangles by (1.20). An argument valid for arbitrary bounded sign sequences must apply to it. No assertion that $pq + h$ is prime enters the construction; the primes are row and column indices. $\square$

5 Exact separated-variable substitutions

We consider (1.21) on Cartesian pieces, with separate row and column maps and nonzero rational weights; no affine or monotonicity assumption is made on the maps. The output-height condition will be imposed only when the integer piece-count bound is deduced.

Proposition 5.1 (Constant multipliers). *Let A, B be finite sets of distinct rational numbers of sizes $r, s \geq 1$, let $d \in \mathbb{Q} \setminus \{0, 1\}$ and $h \neq 0$, and let F, G be rational-valued functions. The matching graph*

$$\mathcal{E} = \{(x, y) : F(x)G(y) + h = d(xy + h)\} \quad (5.1)$$

is $K_{2,2}$ -free, and its edge count satisfies

$$|\mathcal{E}| \leq \min(r + s\sqrt{r}, s + r\sqrt{s}). \quad (5.2)$$

A Cartesian covering by such nonempty pieces requires at least $\min(r, s)$ pieces; with rational outputs this is attained. For arbitrary masks covering density ρ , the consequence is only

$$Q \geq \frac{\rho rs}{\min(r + s\sqrt{r}, s + r\sqrt{s})}. \quad (5.3)$$

Proof. Write $c = (d - 1)h \neq 0$. For distinct row and column pairs,

$$(dx_1y_1 + c)(dx_2y_2 + c) - (dx_1y_2 + c)(dx_2y_1 + c) = dc(x_1 - x_2)(y_1 - y_2) \neq 0. \quad (5.4)$$

The corresponding minor of $F(x)G(y)$ is zero, excluding all four matching edges. For row degrees t_x and $e = |\mathcal{E}|$,

$$\sum_x \binom{t_x}{2} \leq \binom{s}{2}, \quad \frac{e^2/r - e}{2} \leq \frac{s(s-1)}{2}, \quad e \leq \frac{r + \sqrt{r^2 + 4rs(s-1)}}{2} \leq r + s\sqrt{r}. \quad (5.5)$$

Interchange the two sides. A Cartesian piece has a singleton side, hence at most $\max(r, s)$ points. Counting gives $\min(r, s)$ pieces, attained by rows with $F(x_0) = 1$, $G(y) = dx_0y + c$, or by columns. For masks each graph has at most the displayed edge count and the union bound gives (5.3). $\square$

Proof of the classification in Theorem 1.6. The 2×2 minor of $(xy + h)$ is $h(x_1 - x_2)(y_1 - y_2) \neq 0$. In (1.21) the same minor of the right side is $R(x_1)R(x_2)S(y_1)S(y_2)h(x_1 - x_2)(y_1 - y_2) \neq 0$, since the weights are nonzero, and the corresponding minor of the left side is $h(U(x_1) - U(x_2))(V(y_1) - V(y_2))$. Hence this product is nonzero and U, V are injective. Divide the identity by $R(x)S(y)$:

$$\frac{U(x)}{R(x)} \frac{V(y)}{S(y)} + \frac{1}{R(x)} \frac{h}{S(y)} = xy + h. \quad (5.6)$$

For distinct y_1, y_2 the vectors $(V(y)/S(y), h/S(y))$, $y = y_1, y_2$, are independent: their determinant is $h(V(y_1) - V(y_2))/(S(y_1)S(y_2)) \neq 0$ by injectivity of V . Solving this linear system gives rational a, b, c, d with

$$U(x)/R(x) = ax + b, \quad 1/R(x) = cx + d, \quad \Delta = ad - bc \neq 0. \quad (5.7)$$

The two row functions are independent because the matrix has rank two. Compare the coefficient and constant for two distinct x :

$$aV(y) + hc = S(y)y, \quad bV(y) + hd = hS(y), \quad (ha - by)V(y) = h(dy - hc). \quad (5.8)$$

If $ha - by = 0$, the other side vanishes too, forcing $ad = bc$. Therefore denominators are nonzero, $S(y) = h\Delta/(ha - by)$, and (1.22) follows. The converse is the expansion

$$(ax + b)h(dy - hc) + h(cx + d)(ha - by) = h\Delta(xy + h). \quad (5.9)$$

Divide by the two denominators and factor the resulting weight into row and column functions. $\square$

Remark 5.2. Nonconstant rational substitutions do exist. The parameters $a = d = 1, c = 0, b = t$ give

$$U(x) = x + t, \quad V(y) = \frac{hy}{h - ty}, \quad R(x)S(y) = \frac{h}{h - ty} \quad (5.10)$$

where the denominator is nonzero. Integer outputs are essential for the next bound.

Lemma 5.3 (Integer outputs). *Scale a, b, c, d to integers with $\Delta \neq 0$, and suppose integer $h \neq 0$. If $c \neq 0$, at most $2\tau(|\Delta|)$ integer x have $U(x) \in \mathbb{Z}$. If $b \neq 0$, at most $2\tau(|h^2\Delta|)$ integer y have $V(y) \in \mathbb{Z}$. If $b = c = 0$, the substitution preserves products:*

$$U(x) = (a/d)x, \quad V(y) = (d/a)y, \quad U(x)V(y) = xy, \quad R(x)S(y) = 1. \quad (5.11)$$

Proof. For integer outputs u, v , direct multiplication gives

$$(cx + d)(cu - a) = -\Delta, \quad (ha - by)(bv + hd) = h^2\Delta. \quad (5.12)$$

The nonzero denominators are signed divisors of these fixed integers; there are twice the positive divisor counts, and nonconstant denominators are injective functions of the integer input. If $b = c = 0$, $ad \neq 0$, and substitution gives (5.11). For later use, for every $\epsilon > 0$ there is C_ϵ with $\tau(n) \leq C_\epsilon n^\epsilon$ for all positive integers. Indeed for $p \geq 2^{1/\epsilon}$, $k + 1 \leq 2^k \leq p^{\epsilon k}$; for each of the finitely many smaller primes $\sup_{k \geq 0} (k + 1)p^{-\epsilon k} < \infty$. Multiply those finite constants and then the prime-power bounds. $\square$

Proof of the integer piece-count assertions of Theorem 1.6. If a side has at most two elements, take $K \geq 2$. Otherwise each side has at least three, so apply the classification. Choose distinct $x_i \in A$, $i = 1, 2, 3$, with distinct integer $u_i = U(x_i)$. Their equations are

$$ax_i + b - cx_iu_i - du_i = 0 \quad (i = 1, 2, 3). \quad (5.13)$$

The 3×4 integer coefficient matrix has rank three. If its rational kernel had dimension at least two, perturb the known nondegenerate vector v by tw along an independent kernel vector. Avoid the at most two zeros of its determinant and the finitely many values making one of the three denominators zero, and choose nonzero rational t . The two distinct nondegenerate Möbius functions would have the same values at three distinct x_i . Their cross-multiplied quadratic has three roots and is identically zero. Unique factorisation of the coprime linear numerators and denominators, including constant denominators, forces proportional coefficient vectors, contradicting the perturbation.

The signed 3×3 minors give an integer kernel vector; divide by its common gcd. Each matrix entry has modulus at most X^{C+1} and each minor is a sum of six triple products, so

$$\max(|a|, |b|, |c|, |d|) \leq 6X^{3C+3}, \quad 0 < |\Delta| \leq 72X^{6C+6}. \quad (5.14)$$

The kernel is one-dimensional, so this representative defines the same map on all of A and the column map is unchanged by common scaling. If $b = c = 0$ the piece preserves products. Otherwise Lemma 5.3, with $\epsilon = \eta/(6C + 6)$, gives

$$\min(|A|, |B|) \leq 2C_\epsilon \max(72, 72h^2)^\epsilon X^\eta. \quad (5.15)$$

Take K the maximum of this constant and 2; it depends only on h, C, η . Every piece that does not preserve products has at most $KX^\eta \max(r, s)$ points. The union of Q pieces has at most their total size, even with overlaps, proving (1.23). For $r, s \geq X^\delta$ and any fixed $\eta > 0$, $Q \geq \rho X^{\delta-\eta}/K$; in particular $Q = X^{o(1)}$ cannot cover fixed positive density (take $\eta = \delta/2$).

For the upper construction, take one row at a time and set

$$U(x_0) = 1, \quad V(y) = 2x_0y + h, \quad R(x_0) = 2, \quad S(y) = 1. \quad (5.16)$$

Then $UV + h = 2(x_0y + h)$. If $h < 0$, $x_0y > -h$ gives $2x_0y + h > x_0y > 0$; positivity for $h \geq 0$ is immediate. For fixed h and sufficiently large X , $V \leq 2X^2 + |h| \leq X^3$. Moreover $UV - xy = xy + h > 0$, so this piece does not preserve products. Rows give r pieces and columns give s . For $r, s \asymp X^\delta$ the lower bound for every fixed η matches this upper bound in the exponent. The upper construction is asserted with output exponent 3, rather than for every prescribed smaller C . $\square$

6 Nonnegative divisor weights

Expectations in the positive-weight results of this section are in the independent divisibility law: at each p the valuation V_p has distribution $(1 - 1/p)p^{-k}$, $k \geq 0$, and any residue coordinates are independent Haar coordinates. They are not averages on a finite interval of integers.

6.1 The squarefree weight with coefficient 16

Proposition 6.1 (A specified root second moment). *Fix even $h \geq 2$. Let $\mathcal{Q}$ consist of primes at most e^L that are not 1 (mod 5) and do not divide h . On squarefree products of $\mathcal{Q}$ put $a(u) = 16^{\omega(u)}$; set*

$$w(n) = 17^{\omega_{\mathcal{Q}}(n)}, \quad S = \prod_{p \in \mathcal{Q}} (1 + 16/p). \quad (6.1)$$

Let d be a centre label with $\log d \leq 2L$ and take $0 < \eta \leq 1$. Use bins with $j\eta \leq 100L$, $\omega(u) \leq 100 \log L$ and $j\eta \leq \log(du) < (j+1)\eta$. Add an independent uniform coordinate modulo a prime $v > e^{101L}$, $v \nmid hd$, and set

$$\rho_{j,v}(n, d) = \frac{1}{w(n)} \sum_{\substack{u|n, \ u \in \mathcal{Q}^{\text{sf}} \\ \omega(u) \leq 100 \log L, \ j\eta \leq \log(du) < (j+1)\eta}} 16^{\omega(u)} \mathbf{1}_{v|n+hd u}. \quad (6.2)$$

Then, for large L ,

$$\mathbb{E} \left[w \sum_j \rho_{j,v}^2 \right] \gg_h (S/v) L^{-24/17}. \quad (6.3)$$

All unequal-padding off-diagonal terms vanish. An upper bound $S(v^2 L)^{-1} (\log L)^\beta$, with fixed constants, requires $v \ll_h L^{7/17} (\log L)^\beta$.

Proof. Given the $\mathcal{Q}$ configuration, choose two padding divisors independently with probabilities $16^{\omega(u)}/w$. The bins have $u_i \leq e^{101L}$. Two target conditions require $v \mid hd(u_1 - u_2)$; since $(v, hd) = 1$, distinct such labels are incompatible. Equal labels have target probability $1/v$. Under the w -tilted law the equality probability is

$$D_{\mathcal{Q}} = \prod_{p \in \mathcal{Q}} \left(1 - \frac{32}{17(p+16)} \right) \asymp_h L^{-24/17}. \quad (6.4)$$

The prime harmonic density of the three classes other than $1 \pmod{5}$ is $3/4$ by the fixed-modulus prime number theorem; multiplying it by $32/17$ gives $24/17$. Conditional on equality, the label includes each prime independently with probability

$$\frac{256}{17p+240} \leq \frac{16}{p}, \quad \mathbb{E} \log u \leq 17L, \quad \mathbb{E} \omega(u) \leq 17 \log L \quad (6.5)$$

for large L . Markov discards at most $17/98 + 17/100 < 1/2$ of this mass when imposing $\log u \leq 98L$ and $\omega(u) \leq 100 \log L$. These labels lie in the bins since $\log d \leq 2L$. Their fixed positive fraction gives (6.3). Comparing the bounds forces $v \ll_h L^{7/17} (\log L)^\beta$, which fails at $v = X^\theta$, $X = \exp(L^{625})$, fixed $\theta > 0$. An additional source condition $u_0 \mid n$ on a separate uniform coordinate multiplies both moments by $1/u_0$ only when $(u_0, v \prod_{p \in \mathcal{Q}} p) = 1$; no such factorisation is asserted with overlap. $\square$

6.2 Arbitrary positive prime-power coefficients

Lemma 6.2 (Full collision law). *Let $\mathcal{P}$ be a finite set of primes at most $Y \geq 2$. For each prime take finitely supported $a_{p,k} \geq 0$, with $a_{p,0} = 1$, and extend multiplicatively to $a(u)$ supported on $\mathcal{P}$ -smooth integers. Put*

$$w(n) = \sum_{u \mid n} a(u), \quad S = \mathbb{E} w = \sum_u a(u)/u, \quad q_n(u) = a(u) \mathbf{1}_{u \mid n} / w(n). \quad (6.6)$$

Tilt by w/S , denoting this law by $\mathbb{P}^, \mathbb{E}^*$, and choose two divisors independently using q_n . There is an absolute constant $c > 0$, uniform in all coefficients and supports, with*

$$\mathbb{P}^*(U_1 = U_2) \geq c/(\log(2Y))^2. \quad (6.7)$$

Proof. At one prime write $t = 1/p$, $a_k = a_{p,k}$, $w_v = \sum_{k \leq v} a_k$ and $S_p = \sum_k a_k t^k$. The local diagonal is

$$D_p = S_p^{-1} \sum_{v \geq 0} (1-t) t^v \frac{\sum_{k \leq v} a_k^2}{w_v}. \quad (6.8)$$

For unequal exponents $i < j$, use $w_v \geq w_j$ and sum $v \geq j$ to get

$$1 - D_p \leq \frac{2}{S_p} \sum_{j \geq 1} a_j \frac{w_{j-1}}{w_j} p^{-j} \leq \frac{2}{S_p} \sum_{j \geq 1} w_{j-1} p^{-j} = \frac{2}{p-1}. \quad (6.9)$$

The last equality exchanges nonnegative sums. For small primes use $b_k = a_k p^{-k}$ and $w_v \geq a_i + a_j \geq 2\sqrt{a_i a_j}$. The probability that selected exponents differ by at least R is at most

$$\frac{1}{S_p} \sum_{\ell \geq R} p^{-\ell/2} \sum_{i \geq 0} \sqrt{b_i b_{i+\ell}} \leq \frac{p^{-R/2}}{1 - p^{-1/2}}. \quad (6.10)$$

Each inner sum is at most S_p by Cauchy–Schwarz. For $R = 8$ and $p \geq 2$ the upper bound is at most $1/(16(1 - 1/\sqrt{2})) < 1/4$. For a probability vector q ,

$$\sum_{|i-j|<8} q_i q_j \leq 15 \sum_i q_i^2. \quad (6.11)$$

Each of the fifteen diagonals is bounded by the right diagonal via Cauchy–Schwarz. Condition on the valuation and average in the tilted law; this proves $D_p \geq 1/20$ for every prime, including 2, 3. Independence and multiplicative coefficients factor the tilted law and both choices, so equality is equality at every prime and

$$\mathbb{P}^*(U_1 = U_2) = \prod_{p \in \mathcal{P}} D_p \geq c_1 \prod_{5 \leq p \leq Y} \left(1 - \frac{2}{p-1}\right). \quad (6.12)$$

Its logarithm is $-2 \sum_{p \leq Y} 1/p + O(1)$, since the errors are summable. Mertens’ prime harmonic estimate proves (6.7). $\square$

Corollary 6.3 (Full-support root consequence). *If all supported labels are at most U , are included in bins, and a new prime $v > U$ is coprime to hd and $\mathcal{P}$, then the target $v \mid n + hdu$ gives*

$$\mathbb{E}[w \sum_j \rho_{j,v}^2] \geq \frac{cS}{v(\log(2Y))^2}. \quad (6.13)$$

For $Y = e^L$, the upper normalisation $S(v^2 L)^{-1}(\log L)^\beta$ forces $v \leq CL(\log L)^\beta$. This full-support bound is not being transferred to arbitrary truncations.

Proof. Unequal labels have incompatible target residues; equal labels have target probability $1/v$. Use Lemma 6.2 and compare the two bounds. $\square$

Lemma 6.4 (Deterministic retained labels). *In Lemma 6.2, let E be any supported-label subset with $\sum_{u \in E} a(u)/u \geq \delta S$, $0 < \delta \leq 1$ fixed. Uniformly in the coefficients, supports and E ,*

$$\mathbb{P}^*(U_1 = U_2 \in E) \geq \delta \exp\{-(2 \log \log(2Y) + C)/\delta\} \gg_\delta (\log(2Y))^{-2/\delta}. \quad (6.14)$$

Proof. The tilting factor cancels the choice denominator, so the divisor marginal is

$$\pi(u) = \mathbb{P}^*(U_1 = u) = a(u)/(uS). \quad (6.15)$$

It and the joint two-choice law factor over primes. At one prime the conditional collision probability for a selected exponent k with $a_k > 0$ satisfies

$$\phi_p(k) = a_k p^k \sum_{v \geq k} (1 - 1/p) p^{-v} / w_v \geq (1 - 1/p) a_k / w_k. \quad (6.16)$$

Averaging its negative logarithm in b_k/S_p gives

$$\begin{aligned} \sum_k \frac{b_k}{S_p} \log \frac{1}{\phi_p(k)} &\leq -\log(1 - 1/p) + \frac{1}{S_p} \sum_{k \geq 1} a_k p^{-k} \log(1 + w_{k-1}/a_k) \\ &\leq -\log(1 - 1/p) + \frac{1}{p-1}. \end{aligned} \quad (6.17)$$

Zero coefficients have zero probability and are omitted. The last inequality uses $\log(1+x) \leq x$ and the nonnegative-sum identity in (6.9). For $\phi(u) = \mathbb{P}^*(U_2 = u \mid U_1 = u)$, factorisation gives

$$\mathbb{E}_\pi \log(1/\phi(u)) \leq \sum_{p \in \mathcal{P}} \{-\log(1-1/p) + 1/(p-1)\} \leq 2 \log \log(2Y) + C =: H_Y. \quad (6.18)$$

The prime harmonic estimate and summable $O(p^{-2})$ error prove the last bound, treating 2 separately. If $e = \pi(E) \geq \delta$, the conditional mean of the nonnegative logarithm is at most H_Y/e . Jensen gives

$$\mathbb{P}^*(U_1 = U_2 \in E) = e \mathbb{E}_\pi[\phi \mid E] \geq e e^{-H_Y/e} \geq \delta e^{-H_Y/\delta}, \quad (6.19)$$

since $e e^{-H_Y/e}$ is increasing for $e > 0$. This proves the lemma. $\square$

Corollary 6.5 (Retained-bin root consequence). *Add an independent uniform coordinate at a prime $v \notin \mathcal{P}$ and assign each label in E to one bin. For fixed h, d put*

$$\rho_{j,v}(n) = w(n)^{-1} \sum_{\substack{u \in E_j \\ u|n}} a(u) \mathbf{1}_{v|n+hdu}. \quad (6.20)$$

Then

$$\mathbb{E}[w \sum_j \rho_{j,v}^2] \geq \frac{S}{v} \mathbb{P}^*(U_1 = U_2 \in E) \gg_\delta \frac{S}{v(\log(2Y))^{2/\delta}}. \quad (6.21)$$

At $Y = e^L$, (1.25) requires

$$v \ll_{\delta, C} L^{2/\delta-1} (\log L)^\beta. \quad (6.22)$$

It fails for $v = X^\theta$, $X = \exp(L^A)$, with fixed $\theta, A > 0$.

Proof. Keep only the nonnegative diagonal terms. For a fixed label the target has probability $1/v$ in its independent coordinate; no off-diagonal congruence assumption is needed. Apply Lemma 6.4 and compare bounds. $\square$

6.3 Cofactor entropy and vertex-dependent retention

Lemma 6.6 (Cofactor entropy). *Let $a(u) \geq 0$ have arbitrary finite support on $\mathcal{P}$ -smooth integers, $a(1) > 0$, without a multiplicativity assumption. Use w, S, q_n of (6.6). If a measurable retained-edge indicator $B(n, u)$ has probability at least $\delta > 0$ in the tilted and selected joint law, then*

$$\mathbb{E}^* \sum_u q_n(u)^2 \mathbf{1}_{B(n,u)} \geq \delta \exp(-H_{\mathcal{P}}/\delta), \quad (6.23)$$

with $H_{\mathcal{P}}$ as in (1.26).

Proof. Write $K = (v_p(U))_{p \in \mathcal{P}}$ and $V = (v_p(n))_{p \in \mathcal{P}}$. Cancel the tilting and selection denominators. The joint probability at $V = K + J$ is

$$\mathbb{P}^*(U = u, V = K + J) = \frac{a(u)}{uS} \prod_{p \in \mathcal{P}} (1-1/p) p^{-J_p}, \quad J_p \geq 0. \quad (6.24)$$

Thus K has the harmonic marginal and the cofactor J is an independent vector of geometric variables even when a is not multiplicative. Given V , knowing K is equivalent to knowing J . Since q_n depends only on V ,

$$\mathbb{E}^* \log(1/q_n(U)) = H(K \mid V) = H(J \mid V) \leq H(J) = H_{\mathcal{P}}. \quad (6.25)$$

Unit coordinates do not alter the conditional divisor probabilities. The geometric entropy at p is $-\log(1 - 1/p) + \log p/(p - 1)$; summation gives the explicit formula and $H_{\mathcal{P}} \ll \log(2Y)$ from $\sum_{p \leq Y} \log p/p = O(\log(2Y))$ and summable higher-order errors. The entropy inequality is the log-sum inequality, or nonnegativity of mutual information. All entropies are finite: K has finite support and J finite geometric entropy.

The indicator may read additional independent unit coordinates. If its joint probability is $e \geq \delta$, the conditional mean of the nonnegative $\log(1/q_n(U))$ is at most $H_{\mathcal{P}}/e$. Jensen gives

$$\mathbb{E}^*[\mathbf{1}_B q_n(U)] \geq e e^{-H_{\mathcal{P}}/e} \geq \delta e^{-H_{\mathcal{P}}/\delta}. \quad (6.26)$$

Conditioning on n makes the left side exactly that of (6.23). $\square$

Proof of Theorem 1.7. For every $(n_{\mathcal{P}}, u)$, before retention the target $v \mid n + hdu$ has probability $1/v$. Conditioning on it therefore preserves the joint law of $(n_{\mathcal{P}}, u)$; the extra residue is forced by u . The original nonnegative logarithm still has mean at most $H_{\mathcal{P}}$ in this root-conditioned law. The mass condition (1.24) says that retention has probability at least δ there. Repeat Jensen in that law and retain the diagonal terms of each nonnegative squared bin. Since each retained edge belongs to one bin, this gives

$$\mathbb{E}[w \sum_j \rho_{j,v}^2] \geq \delta(S/v) e^{-H_{\mathcal{P}}/\delta}. \quad (6.27)$$

No independence of B is used. The q_n are the original selection probabilities; they are not replaced by a probability renormalised after root conditioning or retention. Compare (6.27) with (1.25) to get (1.26). More generally it fails at $v = X^\theta$, $X = \exp(L^A)$, for fixed $A > 1$ and $\theta > 0$. $\square$

6.4 Local centring and a short inverse-convolution sum

Proposition 6.7 (Local tilted centring). *Let n be Haar-uniform in $\mathbb{Z}_p$, $v_p(a) = 1$ and $0 < r < 1$. Set $B_r(n) = r^{v_p(n)}$. Then*

$$T_p(r) := \mathbb{E}[B_r(n)B_r(n+a) \mid p \mid n] = r^2 \frac{p-2+r}{p-r}, \quad (6.28)$$

$$\mathbb{E}[(\mathbf{1}_{p|n} - 1/p)B_r(n)B_r(n+a)] = \frac{p-1}{p^2}(T_p(r) - 1) < 0, \quad a_p(r) = \frac{T_p(r)}{p-1+T_p(r)} \quad (6.29)$$

is the unique scalar replacing $1/p$ to make the expectation zero. For $C_c(n) = c^{\mathbf{1}_{p|n}}$, $0 < c < 1$, the same formulas hold with $T_p(r)$ replaced by c^2 . A single scalar depending only on p, r does not centre both the first and second powers of the pair weight. The analogous assertion holds for C_c .

Proof. Given $p \mid n$, write $n = pm$ and $a = pb$ with b a unit. There are $p-2$ residue classes where both $m, m+b$ are units, and two where exactly one is divisible. For $j \geq 1$ each valuation-tail probability is $(1 - 1/p)p^{-j}$, with disjoint events. Thus

$$\mathbb{E}[r^{v_p(m)+v_p(m+b)}] = 1 - 2/p + 2(1 - 1/p) \sum_{j \geq 1} (r/p)^j = \frac{p-2+r}{p-r}. \quad (6.30)$$

The two extracted factors give r^2 . Outside $p \mid n$ both weights are one. Separating the two events gives the covariance. The full weighted mean is $(p-1+T)/p$ and its divisible part is T/p , whose ratio is the unique centre. For the Jordan local factor the divisible weight is c^2 , including when $p = 2$. The two expectations that would require a common scalar are

$$\mathbb{E}[(\mathbf{1}_{p|n} - a_p)(B_r(n)B_r(n+a))^q] = 0 \quad (q = 1, 2). \quad (6.31)$$

For the second power the required centre is $a_p(r^2)$. On the divisible event the weight is strictly smaller, so $T_p(r^2) < T_p(r)$ and $T/(p-1+T)$ is strictly increasing. The Jordan centres are $c^2/(p-1+c^2)$ and $c^4/(p-1+c^4)$, also different. This proves only the scalar assertion; a centring depending on a whole word is outside it. $\square$

Remark 6.8. A weight involving all primes reads the centre-prime coordinates as well as the other coordinates. The covariance in (6.29) therefore changes the original singleton centring. For Jordan weights with $\sigma = c/\log X$, $X = \exp(L^A)$ and $p \leq e^L$,

$$(1 - p^{-\sigma})^2 \leq \sigma^2 (\log p)^2 \leq c^2 L^{2-2A}. \quad (6.32)$$

For $A > 1$, the centring correction has size comparable to $1/p$; dividing by the small raw divisible mass does not make it a small relative error. The balanced small-prime soft weight of [12, Theorem 1.3] reads only primes below the centre primes. Its argument does not justify replacing that mask by a full Jordan or full- Ω mask inside the operator while keeping the old centring.

Remark 6.9 (Analytic input for the short sum). The estimate in Proposition 6.10 below, like Remark 2.5, is relative to the zero-free half-plane $\Re s > 7/8$ of OpenAI, family 003 [19]. Here are the short analytic deductions used. Put $a = 7/8$. For every $\delta, \xi > 0$, uniformly over all characters modulo q ,

$$|L(s, \chi)^{-1}| \ll_{\delta, \xi} \{q(|\Im s| + 3)\}^\xi \quad (\Re s \geq a + \delta), \quad (6.33)$$

with analytic value zero at a principal pole. It suffices to prove this for $0 < \delta < 1 - a$. For a primitive nonprincipal character, partial summation of periodic partial sums bounded by q gives $|L(s, \chi)| \leq q|s|/\Re s$ for $\Re s > 0$. Choose the analytic logarithm $F(s) = \log L(s, \chi)$ in the simply connected zero-free half-plane, agreeing with the Euler logarithm on $\Re s > 1$. Borel–Carathéodory on discs centred at $2 + it$ with left edges $a + \delta/4, a + \delta/2$ gives $|F(a + \delta/2 + it)| \ll_\delta \log(q(|t| + 3))$, using the polynomial bound and $F(2 + it) = O(1)$. On $b = 1 + \delta/2$ the Euler logarithm is $O_\delta(1)$. Apply three-lines to $F(s) \exp((s - it_0)^2/R^2)$, $R = q + |t_0| + 3$; its finite boundary suprema give

$$|F(a + \delta + it_0)| \ll_\delta \{\log(q(|t_0| + 3))\}^{1-\delta/(2(1-a))}. \quad (6.34)$$

Exponentiation, and $\exp(C(\log Q)^{1-c}) \ll_{c, \xi} Q^\xi$, give (6.33); to the right of b use the Euler product. For the primitive principal character use instead $Z(s) = \zeta(s)(s-1)/(s+1)$, analytic and nonzero including at 1. The formula $\zeta(s) = s/(s-1) - s \int_1^\infty \{u\} u^{-s-1} du$ bounds Z polynomially, and the same logarithm argument bounds both Z and its reciprocal. Since $|(s-1)/(s+1)| \leq 1$ for $\Re s \geq 0$, the reciprocal zeta bound follows. Imprimitive reciprocal Euler factors have logarithmic modulus at most $C_\delta \sum_{p|q} p^{-a-\delta}$. For any $\xi > 0$ choose a fixed large P so that $C_\delta p^{-a-\delta} \leq \xi \log p$ for $p > P$; small primes cost a constant and $\sum_{p|q} \log p \leq \log q$. Splitting the exponent margin gives the uniform imprimitive bound. Larger δ follow by weakening it.

For Liouville the exact Euler series is

$$\sum_{n \geq 1} \lambda(n) \chi(n) n^{-s} = L(2s, \chi^2)/L(s, \chi) \quad (\Re s > 1). \quad (6.35)$$

The numerator converges absolutely to the right of $a + \delta$, and its principal pole is at $1/2$, outside that region. Perron at height Y^3 and line $1 + 1/\log(2Y)$ has coefficient error $O(1 + Y \log^2(2Y)/Y^3)$: bound the two nearest cutoff coefficients by one and sum $Y/(Y^3|n - Y|)$ for the others. Shift to $a + \delta$ and use (6.33); arbitrarily small exponent margins on the vertical and horizontal integrals give $\sum_{n \leq Y} \lambda(n) \chi(n) \ll_\xi q^\xi Y^{7/8+\xi}$. For $n \equiv r \pmod{q}$, let $g = (r, q)$ and write $n = gm$, reducing

to one unit class modulo q/g . Complete multiplicativity and character orthogonality (coefficient absolute sum one) give

$$\sum_{\substack{n \leq Y \\ n \equiv r \pmod{q}}} \lambda(n) \ll_{\xi} q^{\xi} Y^{7/8+\xi}, \quad \sum_{k \leq Z} \lambda(qk+h) \ll_{h,\xi} q^{\xi} (qZ+h)^{7/8+\xi}. \quad (6.36)$$

The last sum subtracts its fixed prefix through h ; it includes nonunit residues.

Proposition 6.10 (Squarefree inverse and a short Type I sum). *Fix $h \geq 1$. For every $X \geq 1$,*

$$\sum_{md \leq X} \lambda(m) \mu^2(d) \lambda(md+h) = \lambda(1+h). \quad (6.37)$$

Relative to the zero-free input in Remark 6.9, for every $\epsilon > 0$, uniformly for $1 \leq M \leq X^{1/8}$,

$$\left| \sum_{m \leq M} \lambda(m) \sum_{d \leq X/m} \mu^2(d) \lambda(md+h) \right| \ll_{h,\epsilon} X^{15/16+\epsilon} M^{1/2}. \quad (6.38)$$

With $C_h(X) = \sum_{n \leq X} \lambda(n) \lambda(n+h)$ this gives

$$C_h(X) = - \sum_{M < m \leq X/2} \lambda(m) \sum_{2 \leq d \leq X/m} \mu^2(d) \lambda(md+h) + O_{h,\epsilon}(X^{15/16+\epsilon} M^{1/2} + M + 1). \quad (6.39)$$

No bound for the remaining bilinear sum is asserted.

Proof. At p^k , $k \geq 1$, the convolution coefficient is $\lambda(p^k) + \lambda(p^{k-1}) = 0$; at 1 it is one. Thus $\lambda * \mu^2 = \mathbf{1}_{\{1\}}$ by multiplicativity. Summing its coefficient at md against $\lambda(md+h)$ proves (6.37). This uses the value $\lambda(p) = -1$ at every prime. For fixed m expand $\mu^2(d) = \sum_{a^2|d} \mu(a)$, and choose $A = \max(1, \lfloor X^{1/16} m^{-1/2} \rfloor)$; the unfloored expression is at least one on the stated range. For $a \leq A$, (6.36) with $q = ma^2 \leq X$, reducing its exponent margin if necessary, gives total $O_{h,\epsilon}(AX^{7/8+\epsilon})$. For $a > A$ the term-count bound is

$$\sum_{a > A} \left\lfloor \frac{X}{ma^2} \right\rfloor \leq \frac{X}{mA}. \quad (6.40)$$

Both costs are $O_{h,\epsilon}(X^{15/16+\epsilon} m^{-1/2})$ by floor comparison. Sum over $m \leq M$ using $\sum m^{-1/2} \leq 2\sqrt{M}$. Separate $d = 1$ in (6.37) and then $m \leq M$; the removed small- m diagonal costs $O(M)$, and the remaining terms have $d \geq 2$, $m \leq X/2$. This proves (6.39). For $M = X^{\theta}$, fixed $\theta < 1/8$, (6.38) saves a power with small enough ϵ . At $M = X^{1/16}$ a log-squared saving for the displayed remaining sum would imply one for $C_h(X)$, but that additional bound is not proved. An absolute bound costs $O(X \log X)$; the balancing cutoff loses its saving for $m > X^{1/8}$. The exact hyperbolic identity itself does not supply an independent estimate for the isolated correlation. $\square$

6.5 Root spacing and the unchanged CRT comparison

Proposition 6.11 (One-source geometry). *Let v be a positive integer and let an integer interval I contain fewer than v consecutive integers. A bipartite graph with source vertices $n \in I$, $v \mid n$, and arbitrary target vertices in I has at most one source. On its nonzero component its weighted adjacency matrix and norm are*

$$A = \begin{pmatrix} 0 & c^* \\ c & 0 \end{pmatrix}, \quad \|A\|_{\text{op}} = \|c\|_2. \quad (6.41)$$

For real coefficients the absolute pairing with source value one and arbitrary target signs can be $\sum_j |c_j|$. No assertion is made that Liouville attains those signs.

Proof. There is at most one multiple of v in I . Squaring the displayed matrix, or computing its two nonzero eigenvalues $\pm \|c\|_2$, gives the norm. Choose the sign of each real c_j for the pairing. $\square$

Remark 6.12 (Scale comparison). For fixed C and $\theta > 0$, $H \leq e^{CL}$ and $v = X^\theta$, $X = e^{L^{625}}$, give

$$H/v \leq \exp(CL - \theta L^{625}) \longrightarrow 0. \quad (6.42)$$

Thus root-source restrictions in these blocks have the one-source geometry, and do not enter a short-progression variance regime requiring $H/v \geq 1$. Larger blocks $H \geq v$ remove this spacing issue. For the label supplies of [12, Definition 2.6 and Proposition 2.7], fix $W \geq 10$, put $\alpha = 1/(1200W)$ and $J = \lfloor \alpha \log L \rfloor$. Each supply has harmonic mass at least W and primes at least $\exp(L^{.995})$, hence at least $W \exp(L^{.995})$ primes. If one keeps a finite comparison requiring uniform marginals for $2kJ$ selected centre coordinates, with $k \asymp \log v$, supplies of at least $W \exp(L^{.995})$ primes allow such selections. Every prime is at least $\exp(L^{.995})$, so their CRT modulus D satisfies

$$\log D \geq 2kJL^{.995} \gg \log X. \quad (6.43)$$

The unchanged full-coordinate CRT bound $D/|I|$ cannot be small on these coordinate sets at origin length X . This comparison uses only the supply and size bounds in the label construction of [12, Definition 2.6 and Proposition 2.7]. It does not say that every such set occurs in a nonzero closed word, or that a specialised event comparison or every longer-block analysis fails.

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